

Symptomatology and investigations of liver diseases

Symptoms of chronic liver disease

o General failure of health

- The most conspicuous feature is **easy fatigability**.
- Muscle wasting, anorexia and malnutrition.

o Jaundice

- Yellowish discoloration of the skin, sclera & mucous membrane, bilirubin $>3\text{mg}$.
- Due to failure of the liver cells to metabolize bilirubin.

o Fever and septicemia

- Patients with decompensated cirrhosis show a continuous low-grade fever which rarely exceeds 38°C .
- **Septicemia** is frequent in terminal hepato-cellular failure.

o Hepatic Encephalopathy

- **Neuropsychiatric** abnormalities occurs in patients with acute or chronic liver disease.

o Ascites

- Free fluid within the peritoneal cavity.
- It forms because of conditions:
 - 1- directly involving the peritoneum (infection, malignancy)
 - 2- diseases remote from the peritoneum (**liver disease**, heart failure, hypoproteinemia).

o Skin manifestations

- **Palmar erythema**: a persistent redness of the palms.
- **Spider naevi**: a small red marks on the skin caused by dilated blood vessels (capillaries) near the surface of the skin.

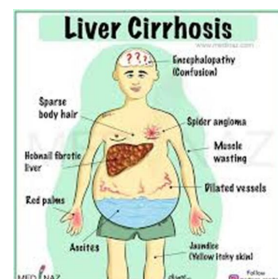
o Endocrine changes

- As gynecomastia and hypogonadism.

o Disordered blood coagulation

- Bleeding tendency:

Liver is the site of synthesis of fibrinogen, factors II, V, VII, IX, XI, XII, and XIII.



o Vasodilation and hyperdynamic circulation

- It is shown by flushed extremities, bounding pulses and capillary pulsations
- Portal blood flow is **increased**.
- The blood pressure is **low**.
- Systemic vascular peripheral resistance is reduced.
- The nature of the vasodilators concerned remains speculative.
- They are likely to be multiple; The substances might be
 - 1) Formed by the sick hepatocyte.
 - 2) Fail to be inactivated by it.
 - 3) Bypass it through intra or extra-hepatic portal-systemic shunts.
- This activates the **sympathetic and renin-angiotensin systems** and is important in sodium and water retention and ascites formation.
- One of the implicated substances in that respect is **Nitric Oxide (NO)**.
- This endothelium-derived potent vasodilator may be involved in the hyperdynamic circulation.
- **NO** is important in ascites formation and the hepato-renal syndrome and in portal hypertension.
- **Prostaglandins (E1, E2 and E12)** have vasodilatory actions and prostanoids are released into the portal vein in patients with chronic liver disease and they may play a part in vasodilatation.

- Cardiovascular Alterations:

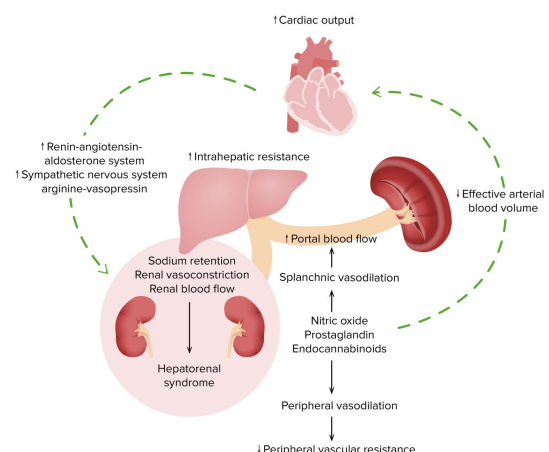
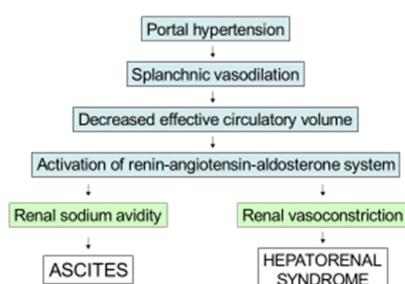
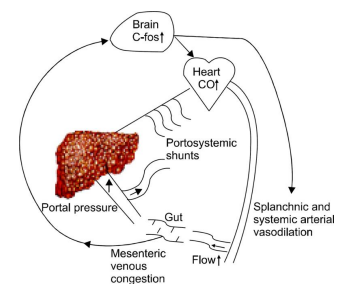
Hepato-pulmonary syndrome.

Porto pulmonary hypertension.

Cirrhotic cardiomyopathy.

- Hepato-renal syndrome:

Is a condition of acute kidney injury seen in patient with advanced liver disease with **normal kidney histology**.



Imaging techniques

Ultrasound examination (US)

- The first imaging modality used to investigate liver disease.
- Non-invasive, safe cheap technique.
- The normal liver appears as a relatively homogeneous structure.
- Color Doppler will demonstrate vascularity within a lesion.



Computed tomography (CT)

- **Triphasic CT** is the investigation of choice for the diagnosis of hepatocellular carcinoma.



Magnetic resonance imaging (MRI)

- o MRI is the most sensitive investigation of focal liver disease
- o **Magnetic resonance cholangiopancreatography (MRCP)** is usually the next test if a biliary abnormality is present on US examination.

Plain X-ray: rarely used but may show gallstones or pancreatic calcification

Fibro-scan:

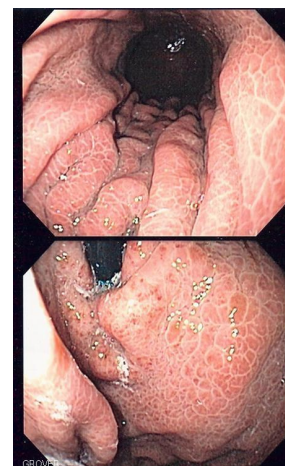
- o Is a special ultrasound technology that measures liver stiffness (hardness) and **fatty changes** in the liver (sensitivity and specificity 80-95% compared to liver biopsy).

Upper GI endoscopy

- o Is used for diagnosis and treatment of varices, detection of portal hypertensive gastropathy, associated lesions such as peptic ulcers.



Esophageal Varices



Portal hypertensive
gastropathy

Endoscopic Retrograde Cholangiopancreatography (ERCP)

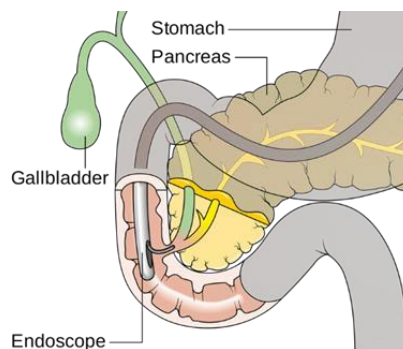
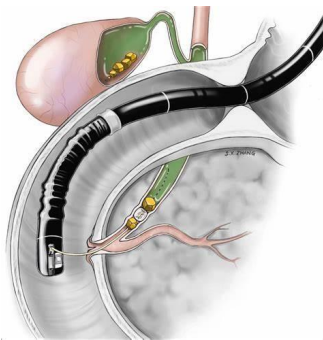
This technique outlines the biliary and pancreatic ducts.

It involves the passage of an endoscope into the **second part of the duodenum** and **cannulation** of the ampulla.

Contrast is injected into both systems and the patient is screened Radiologically.

Therapeutic ERCP involves the following:

- Removal of common bile duct stones.
- Drainage of the biliary system by a stent through an obstruction.



Endoscopic ultrasound (EUS)

A less invasive method for bile duct imaging.

EUS-guided **fine-needle aspiration** of tumors e.g. pancreatic tumors.

Percutaneous transhepatic cholangiography (PTC):

A fine needle is passed into the liver, contrast is injected until a biliary radicle is identified PTC is performed if ERCP fails.

Liver biopsy:

Histological examination of the liver is valuable in the differential diagnosis of diffuse or localized parenchymal disease.

Investigations of liver diseases

Liver plays many important functions in the body as regard carbohydrate, lipid and protein metabolism.

Liver is the site of synthesis of albumin, all clotting factors (except factor VIII), carrier proteins (As ceruloplasmin, transferrin), cholesterol, bile salts and urea.

It also has a detoxification function (removal of ammonia and other toxic substances).

Laboratory diagnostic findings of liver diseases

The most important liver function tests include the following:

1- Albumin and other proteins:

Albumin is the most important serum protein synthesized by the liver.

It has a long half-life (21 days) and is therefore not a good indicator of acute liver injury.

Furthermore, there is a high reserve of hepatic albumin synthesis and thus albumin is decreased only in **chronic liver diseases** with severe cells destruction.

Reduction of serum albumin provides an excellent indicator of severity and prognosis of chronic liver disease.

As regards immunoglobulins, there is a polyclonal increase of gamma globulin (due to ↓ clearance or ↑ synthesis) in chronic liver disease with reversed albumin/globulin ratio.

In liver cirrhosis IgA is mainly ↑ and to a less extent other immunoglobulins. In chronic hepatitis there is a main ↑ in IgG. Total protein in chronic liver diseases is not of diagnostic value (may be ↑ or ↓ or normal).

2- Liver enzymes:

I- Aminotransferases:

They include alanine aminotransferase (ALT or SGPT) and aspartate aminotransferase (AST or SGOT).

These enzymes are present in the hepatocytes and thus their elevations indicate liver cell damage.

ALT is found primarily in the liver and more specific to the liver so that a rise only occurs with liver diseases.

AST is also present in heart, muscle, kidney and brain, High levels are seen in hepatic necrosis, myocardial infarction and muscle injury.

In hepatocytes ALT is found exclusively in cytoplasm while AST exists in both cytoplasm and mitochondria.

In acute viral hepatitis, there is marked elevation of ALT and AST (10-100 times).

The rise of ALT is usually more than AST (ALT/AST ratio >1) because more damage of cytoplasm and slower clearance of ALT.

Serum AST has a shorter half life (17 hour) than ALT (47 hour). Hence, the serum ALT level takes a longer time to normalize than the AST after the injury has subsided.

In liver cirrhosis, there is mild to moderate elevation of ALT and AST (2-5 folds).

The rise of AST is more than ALT (ALT/AST ratio is <1) because the damage affecting the mitochondria of liver cells with release of more AST.

In chronic active hepatitis, there is also mild to moderate elevation of ALT and AST with ALT/AST ratio >1.

II-Alkaline phosphatase (ALP) and gamma glutamyl transferase (GGT):

These enzymes are found in biliary tract epithelium.

In cholestasis (**obstructive jaundice**) and space occupying lesions in liver (As in cancer), there is marked elevation of these enzymes (more than 3 folds).

This marked rise is due to increase enzyme synthesis by cells lining biliary canaliculi.

Slight to moderate elevation (1-2-fold) occur in patients with other liver diseases as **hepatitis** and **cirrhosis**.

GGT is more specific for liver disease than ALP.

The latter (ALP) is also elevated in non-hepatic disorders as bone diseases (rickets ,osteomalacia).

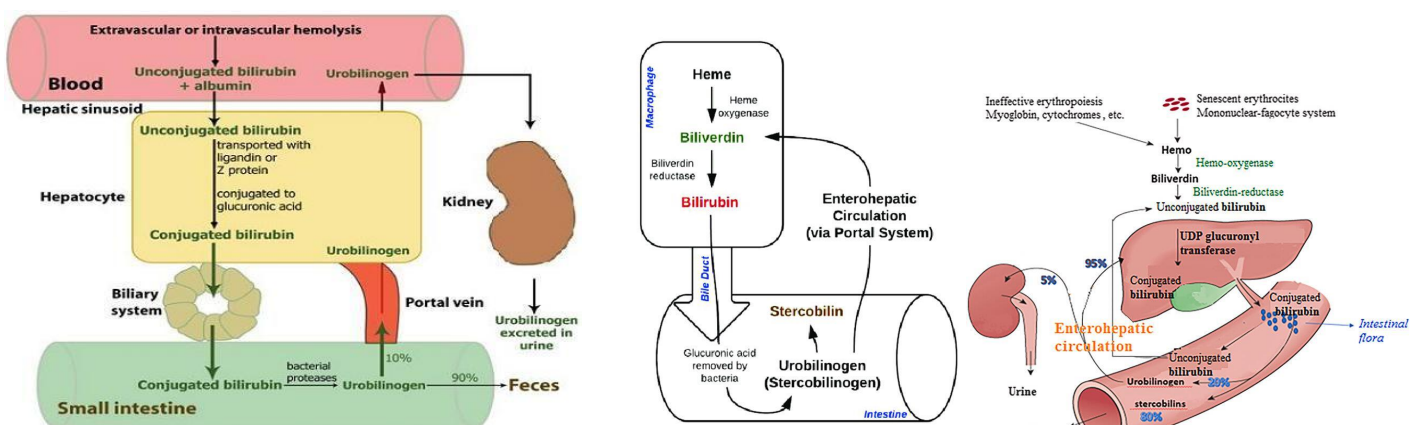
GGT also provides a more sensitive index of cholestasis than ALP.

Elevated levels of GGT are also seen in alcoholic even with normal liver.

Another enzyme called 5'nucleotidase is more specific for cholestasis than ALP but its use is not popular.

3-Bilirubin

- Bilirubin is the end product of heme degradation.
- Heme (By heme oxygenase enzyme) → Biliverdin (By biliverdin reductase) → Bilirubin.
- Hepatocellular processing of bilirubin involves:
 - **Carrier mediated uptake:** by hepatic cells.
 - **Conjugation:** to glucuronic acid by uridine diphosphate- glucuronosyl transferase
 - **Excretion** into bile.
- **Urobilinogen: colorless**
 - Due to deconjugate most of bilirubin by gut bacteria B-glucuronidases.
 - Is largely excreted in feces as stercobilinogen and stercobilin.
 - 10-20% of urobilinogen are reabsorbed in the ileum and colon and escape enterohepatic circulation and pass to the kidney to be excreted in urine.
- Bile acids (cholic&chenodeoxy cholic acids) are derived from cholesterol and secreted in bile.
- Most of them are reabsorbed, especially in the ileum and returned to the liver for re-secretion again (enterohepatic circulation).
- **Unconjugated = indirect bilirubin:** insoluble in water and is tightly complexed to serum albumin and not excreted in urine.
- **Conjugated = direct bilirubin:** water soluble, non toxic, and only loosely bound to albumin and excreted in urine.
- The rate of systemic bilirubin production equal to the rate of hepatic uptake, conjugation and biliary excretion.
- Jaundice occur when the equilibrium between production and clearance is disturbed e.g. excessive production, reduced hepatic uptake, impaired conjugation, decreased hepatocellular excretion and impaired bile flow.



Clinical Applications

1- Unconjugated hyperbilirubinemia unconjugated (indirect) bilirubin level is usually >80% of total bilirubin.

I-Over production as in **hemolytic anemia (hemolytic jaundice)**, **Ineffective erythropoiesis.**

II-Decrease uptake as in **Gilbert syndrome.**

- This syndrome is relatively common and has a benign course.
- Congenital defect of enzymes or binding proteins
- It results from defect of hepatic uptake and/or decrease of UGT activity (with decreased conjugation also).
- Bilirubin level is increased but rarely exceeds 5 mg/dl.
- Liver enzymes are normal, with no evidence of hemolysis (normal reticulocyte count, no hemolytic anemia...).

III- Decrease conjugation

as in **Physiologic jaundice** of newborn and **Crigler-Najjar syndrome**

- Crigler-Najjar syndrome type1:

- The severe form of this syndrome is characterized by complete absence of UGT (uridine diphosphate glucuronosyl transferase) which is the enzyme responsible for bilirubin conjugation.
- It might be fatal within 18 months of birth.
- Total Bilirubin is more than 20 mg/dl.

- Crigler-Najjar syndrome type 11:

Partial deficiency of UGT.

Bilirubin less than 20 mg/dl.

2- Conjugated hyperbilirubinemia, Conjugated (direct) bilirubin level is usually > 50% of total bilirubin:

A- Impaired secretion: Defect of bilirubin transfer into bile.

- Dubin Jonson syndrome: hereditary disorder of bilirubin secretion, Liver is darkly pigmented.

- Rotor Syndrome: usually asymptomatic, Liver is not pigmented.

B- Cholestasis: due to intrahepatic or extrahepatic obstruction

- Intrahepatic due to: hepatic malignancy or infiltration, primary biliary cirrhosis , biliary atresia, cholangitis and drugs (chlorpromazine and some steroids)
- Extrahepatic: due to **gallstones** and **cancer head of pancreas** and external pressure by lymph nodes or tumors.

3- Conjugated and unconjugated hyperbilirubinemia (Mixed) (hepatic jaundice):

- o **Both types of bilirubin are increased** (direct and indirect).
- o It is due to hepatocellular diseases acute viral hepatitis, cirrhosis, hepatic failure, toxic hepatitis due to some drugs (halothan and paracetamol).
- o Liver enzymes are increased.
- o **Urine is dark** due to the presence of direct bilirubin. Urobilinogen increases in early stage and then decrease.

Hemolytic jaundice

- Increase mainly **indirect bilirubin**.
- **Urine is normal** in color: Absent bilirubin, because indirect type is not excreted in urine) , increase urobilinogen in urine.
- **Stool is dark**: increase stercobilin.
- Normal liver enzymes only slight ↑ of AST (present in RBCs).
- Hemolysis: **↑reticulocytes, increased LDH**.

Obstructive jaundice

- Increase mainly **direct bilirubin**.
- **Urine is dark**: positive for bilirubin and bile salts and negative urobilinogen.
- **Stool is clay color**.
- **Marked rise of GGT and ALP**.
- Normal ALT and AST except in late stages with secondary liver cell damage from bile.
- Urobilinogen is decreased or absent.
- prothrombin time (PT) is **prolonged and can be corrected by Vit K injection (mal absorption of fat soluble Vit)**.
- Increase cholesterol and B lipoproteins.
- Pruritus, itching and skin xanthoma are common

Hepatic jaundice

- The laboratory findings are similar to that mentioned in viral hepatitis with increase of both types of bilirubin (direct and indirect).
- Dark urine (positive for bilirubin).

Findings	Hemolytic jaundice	Obstructive jaundice	Hepatic jaundice
S. Bilirubin	↑ indirect	↑ direct	↑ both
Urobilinogen	↑	↓	↑ in early stages, then ↓
Urine bilirubin	absent	Present	Present
Urine bile salts	absent	Present	Present
Urine color	Normal	Dark	Dark
Stool color	Dark	Clay	Clay

Table 17.4 Some causes of obstructive jaundice or cholestasis

Dilated ducts/mechanical obstruction (extrahepatic)	Undilated ducts/non-mechanical obstruction (intrahepatic)
Gallstones	Primary biliary cirrhosis
Pancreatic carcinoma	Primary sclerosing cholangitis
Pancreatic duct stricture	Cholestatic drug reaction
Cholangiocarcinoma	
Parasite infections	

4- Prothrombin time (PT)

The liver is the major site of synthesis of coagulation factors.

PT is dependent on normal hepatic synthesis of coagulation factors (I, II,V,VII and X).

Many of these factors are vitamin K dependant (factor II, VII,IX,X).

Prolongation of PT in liver disease may be due to impaired synthesis of clotting factors or impaired absorption of vitamin K.

In the latter PT is normalize after parenteral injection of vitamin K.

Prolonged Prothrombin time

1- Impaired synthesis of clotting factors:

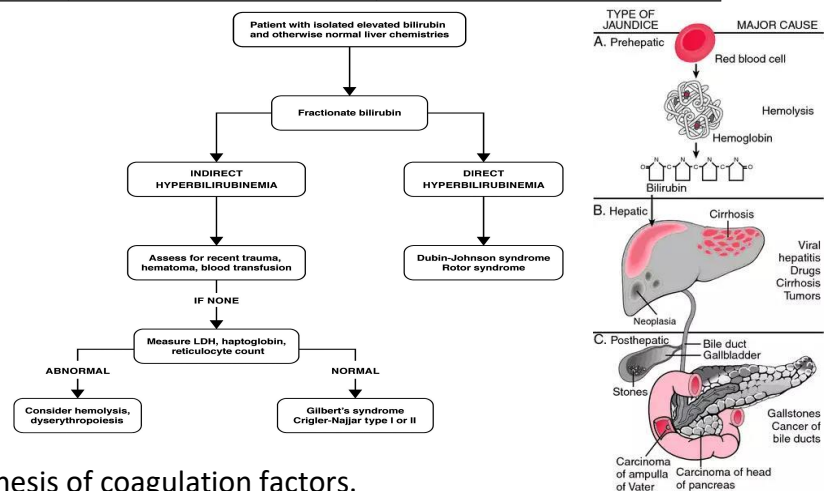
As in chronic liver diseases (cirrhosis).

Because these factors have a shorter half life than albumin PT may be an earlier indicator of severe liver injury than albumin and is useful to assess severity and prognosis of chronic liver disorders.

2- Impaired absorption of Vit K:

Due to obstructive jaundice and thus there is failure of Vit K absorption.

This condition can be distinguished from impaired hepatic synthesis by re-assay of PT after parenteral injection of Vit K, if it is normalize this indicate cholestasis and not decrease liver synthesis



Additional blood investigations:

Viral markers: including hepatitis A, B, C, D, and E.

Biochemical tests:

- α -Fetoprotein: increased in hepatocellular carcinoma , slightly raised in patients with hepatitis and chronic liver disease.
- α 1Anti-trypsin deficiency can produce cirrhosis.
- Raised urinary copper, low serum copper and ceruloplasmin occur in **Wilson's disease**.

Immunological tests:

• Serum immunoglobulins:

- o In primary biliary cirrhosis, raised **AMA**.
- o Autoimmune hepatitis, raised **IgG total**.
- o In liver cirrhosis, **IgA** is mainly increased.
- o In chronic hepatitis, there is a main increase in **IgG**.

• Serum auto-antibodies:

• Anti-mitochondrial antibodies (AMA's)

Found in 95% of patients with primary biliary cirrhosis (PBC), and are also found in autoimmune hepatitis.

• Anti-nuclear (ANA), anti-smooth muscle (ASMA), liver/kidney microsomal (LKM) antibodies

Increased in autoimmune hepatitis.

• Anti-nuclear cytoplasmic antibodies (ANCA)

Found in primary sclerosing cholangitis.

• Genetic analysis: e.g., haemochromatosis "HFE gene".

Wilson disease

- Wilson disease is a rare autosomal recessive inherited disorder of copper metabolism that is characterized by excessive deposition of copper in the liver, brain, cornea, and other tissues.
- The genetic defect, localized to chromosome 13, has been shown to affect the copper-transporting adenosine triphosphatase (ATPase) gene (ATP7B) in the liver.
- ATP7B mediates the transfer of copper to apoceruloplasmin to form a six-copper binding protein known as ceruloplasmin, which is an α_2 -globulin. ATP7B mediates also excretion of copper into biliary canaliculi.
- Around 50%-75% of intestinal copper is absorbed and then transported to the hepatocytes.
- After copper reaches the hepatocyte, it is incorporated into copper-containing enzymes and copper-binding proteins (CBPs), including ceruloplasmin. Excess copper is excreted into bile.
- In Wilson disease (WD), the processes of incorporation of copper into apoceruloplasmin and excretion of excess copper into bile are impaired.
- The transport of copper by the copper-transporting ATPase is defective in Wilson disease secondary to mutations in the ATP7B gene.
- Mutations in ATP7B lead to a reduction in the conversion of apoceruloplasmin into ceruloplasmin, which, as a result, is usually present at low levels in WD patients.
- In addition, a failure to excrete copper into the biliary canaliculi leads to its toxic build-up within the hepatocytes. Initially, the excess copper accumulates in the liver, leading to damage to hepatocytes.
- Eventually, as liver copper levels increase, it diffuses in the circulation and is deposited in other organs.
- Copper accumulates in the liver, brain, eyes, and other organs.

Clinical picture

- Symptoms usually begin between ages 5 and 35 years.

1- CNS:

The first symptoms result from brain damage.

They include tremors, difficulty speaking and swallowing, in-coordination, involuntary jerky movements (chorea), personality changes, and even psychosis (such as schizophrenia or manic-depressive illness).

2- Liver: there is liver damage due to inflammation (hepatitis) and eventually scarring (cirrhosis).

3- Eye:

Gold or greenish gold rings (Kayser-Fleischer rings) may appear around the irises (the colored part of the eye) or the corneo-scleral junction, may be golden brown, or brownish green or brownish-yellow rings also.

These rings develop when copper accumulates.

In a few people, these rings are the first sign of Wilson disease.

4- Others: Hemolytic anemia due to the toxic effect of copper on red blood cells.

Diagnosis

Doctors suspect Wilson disease based on symptoms, such as hepatitis, tremors, and personality changes.

The following tests help confirm the diagnosis:

Slit-lamp examination of the eye: Kaiser-Fleischer rings.

Blood tests: measure ceruloplasmin (a copper-carrying protein) is low.

Serum ceruloplasmin (normally 20 to 35 mg/dL is usually low in Wilson disease).

Urine: Measurement of copper excreted, 24-hour urinary copper excretion (normally, ≤ 30 mcg/day) is increased and usually > 100 mcg/day.

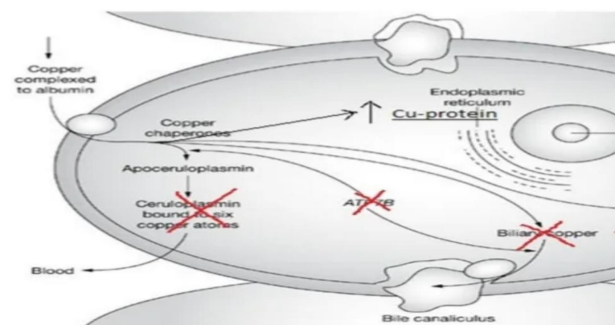
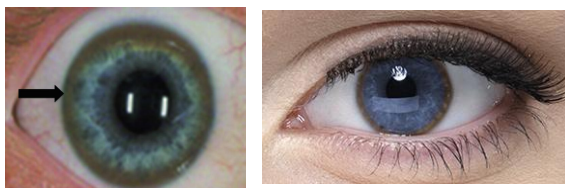
If serum ceruloplasmin is low and urinary copper excretion is high, diagnosis is clear

Liver biopsy: If the diagnosis is still unclear, (indeterminate values for ceruloplasmin and urinary copper), the diagnosis is made by doing a liver biopsy to measure hepatic copper concentration. A diagnosis of Wilson disease is established if the hepatic copper concentration is ≥ 250 mcg/g dry weight.

A diagnosis of Wilson disease is established if the hepatic copper concentration is ≥ 250 mcg/g dry weight.

Serum copper levels are sometimes measured; however, they may be variable.

In general, Total serum copper (which includes copper incorporated in ceruloplasmin) in WD is usually decreased in proportion to the decreased, ceruloplasmin in the circulation but serum non-ceruloplasmin bound copper concentration is elevated.



Diagnosis of Acute viral hepatitis:

1- pre-icteric stage:

- ALT is increased early and is more sensitive and specific than AST.
- Bilirubin is still normal.

2- Icteric stage:

- Marked elevation (10-100 times) of ALT and AST and ALT/AST ratio is usually >1 .
- There is Increase both direct and indirect bilirubin (few cases are Anicteric, and bilirubin is normal).
- ALP and GGT showed mild elevations.
- Serum albumin is normal.
- PT is prolonged in severe cases.
- Urine: is dark brown (Positive for bilirubin and bile salts).

2- Icteric stage:

Marked elevation(10-100 times) of ALT and AST and ALT/AST ratio >1 due to slower clearance of ALT.

3- Convalescent and recovery:

Symptoms and icterus resolve.

Liver enzymes start to decline and return to normal.

Acute viral hepatitis usually resolve 4 to 8 weeks after symptom onset.

Immunological (serological) Diagnosis.

Follow-Up of Acute Hepatitis

Important uses of laboratory tests in acute hepatitis are to identify individuals with fulminant hepatic failure, and to document recovery and clearance of any infectious agents.

The most important tests in determining the extent of injury are not plasma activities of cytosolic enzymes, but evidence of impaired liver function.

The most important indicator of prognosis in acute viral hepatitis is impairment in synthetic function, with PT a widely accepted indicator.

In acute viral, a PT of more than 3 seconds above normal is associated with a poor prognosis.

With hepatitis B and C, cytosolic enzyme activities may return to normal even if viral replication persists; Serologic tests are the only reliable means to evaluate resolution of infection

Immunological (serological) Diagnosis of different types of viral hepatitis:

Hepatitis

Causes

- Viral (Hepatitis A, B, C, D, E, F, G;
- Rarely Epstein-Barr, Herpes, CMV
- Autoimmune
- Drug/toxin induced
- Alcoholic

Hepatitis A:

It is a single stranded RNA picornavirus.

Incubation period is 2-6 week.

Usually benign, self-limited disease. Rarely causes fulminant hepatitis; and never causes chronic hepatitis.

Mode of transmission is Fecal/oral.

Rarely causes fulminant hepatitis; never causes chronic hepatitis.

Endemic in places with poor sanitation.

Hepatitis A virus(HAV) antibodies:

Two types of antibodies: first, IgM, which appears 2 to 4 weeks after infection and is present for 2 to 6 months, then IgG, which appears a few weeks after IgM and is present for the rest of the person's life.

IgG provides long-term immunity (protection), whether due to a prior infection by the virus or due to vaccination.

The presence of Hepatitis A virus IgM diagnoses acute or very recent hepatitis, HAV IgG (old infection).

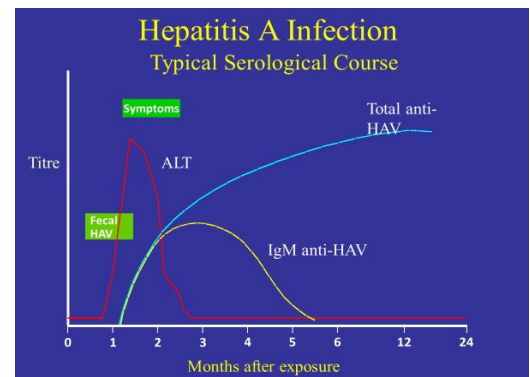
Testing HAV total antibodies(IgG plus IgM): This assay measures both IgG and IgM antibodies, but does not differentiate between them. It may be useful in confirming an individual's long-term immunity and the appropriateness of vaccination.

HAV IgM (recent infection), A positive HAV-Ab total test (without isolated testing for IgM) indicates prior HAV exposure but does not differentiate current from previous HAV infection.

A negative HAV-Ab total test means there has been no prior exposure to the virus and the hepatitis A vaccination should be administered

Direct Detection:

- PCR of feces to detect HAV. Can detect illness earlier than serology but is very rarely performed.



Hepatitis B:

It is Enveloped double stranded-DNA virus.

Incubation period is 2-6 months.

Chronic hepatitis occurs in 5-10% of cases, **It has increased risk of hepatocellular carcinoma.**

Mode of transmission is parenteral (as contaminated blood transfusion or contaminated syringes), sexual or perinatal from infected mother to their newborn.

Present in all body fluids except stool, so spread is by blood or body fluid contact.

Worldwide carrier rate of 300 million.

May coexist with Hepatitis D (delta virus).

Hepatitis B serological markers include:

Hepatitis B surface antigen (HBsAg): Is the earliest marker, appears in the late days of incubation period and denotes infectivity.

Hepatitis B surface antibody (HBsAb or anti-HBs):

The presence of anti-HBs is generally interpreted as **recovery** (the acute infection is completely resolved) and immunity from hepatitis B virus infection.

Anti-HBs also develops in a person who has been successfully **vaccinated** against hepatitis B.

HBsAb does not develop in those with chronic hepatitis, Those who are immune due to natural infection will have HBsAb and HBcAb, Immunity due to vaccination will have only HBsAb.

Hepatitis B core antibody (HBcAb or anti-HBc):

Appears at the onset of symptoms in acute hepatitis B and **persists for life.**

This is the only positive marker in window period (gap between disappearance of HBsAg and the appearance of HBsAb).

The anti-HBc total antibodies test detects both IgM and IgG antibodies, thus a positive result indicates either acute, chronic, or past infection.

Two types are measured in blood:

First the anti-HBc total antibodies test which detects both IgM and IgG antibodies, thus a positive result indicates acute, chronic, or past infection.

Second is IgM antibody to hepatitis B core antigen (HBcAb IgM or IgM anti-HBc): Its positivity indicates recent or acute infection with hepatitis B virus, It is detected within 2 weeks after HBsAg and persists for 3-6 months.

Hepatitis B core antigen is **not detected in blood.**

Hepatitis B markers in window period: gap between disappearance of HBsAg and the appearance of HBsAb .

Usually denotes resolving acute infection, but the infection is not completely resolved because HBsAb has not yet detected.

Anti-HBc is usually the only marker detected in window period

HBsAg	negative
anti-HBc	positive
anti-HBs	negative

The most important markers for diagnosis of **recent infection** are **HBsAg** and **HBcAb IgM**.

Hepatitis B envelop antigen (HBeAg): it is not found in every case. If positive this denotes rapid viral multiplication and high infectivity.

Hepatitis B e antibody (HBeAb): is not found in every case. It appears after disappearance of HBeAg.

HBV DNA by PCR: hepatitis B virus DNA (HBV DNA) is now the most important and reliable marker for monitoring viral replication.

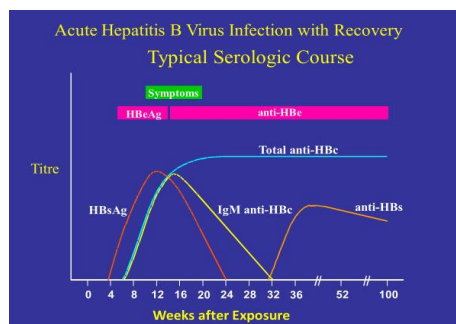
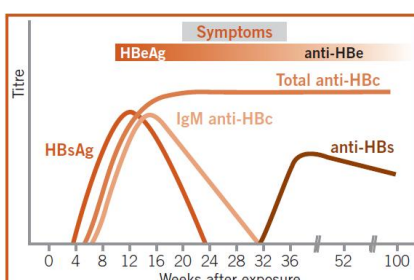
The main advantages of PCR are its extreme sensitivity and quantitative PCR assays help to monitor viral replication and response to treatment.

It can:

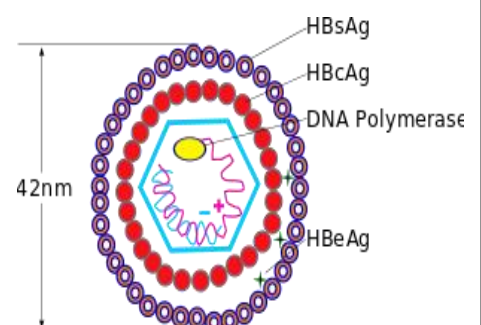
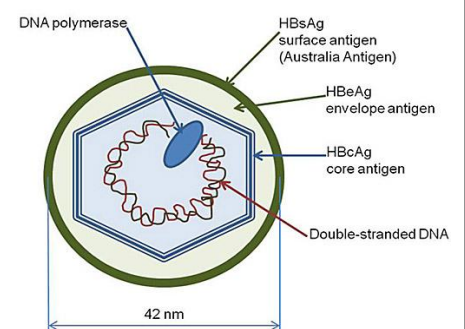
- 1- Diagnose some cases of early acute HBV infection (before the appearance of HBsAg).
- 2- Distinguish active from inactive HBV infection.
- 3- Monitor a patient's response to anti-HBV therapy.

Table 1. Interpretation of hepatitis B markers.

Marker	Acute infection	Chronic infection	Past infection
HBsAg	+	±	-
HBeAg	+ early, then -	±	-
anti-HBs	-	-	+
anti-HBcIgM	+	-	+
anti-HBcIgG	+	+	+
anti-HBe	- early, then +	±	+
HBV DNA	+ early, then -	±	-
ALT	increased (marked)	increased (mild-moderate)	normal



HBsAg	negative	Susceptible
anti-HBc	negative	
anti-HBs	negative	
HBsAg	negative	Immune due to natural infection
anti-HBc	positive	
anti-HBs	negative	
HBsAg	negative	Immune due to hepatitis B vaccination
anti-HBc	negative	
anti-HBs	positive	
HBsAg	positive	Acutely infected
anti-HBc	positive	
IgM anti-HBc	positive	
anti-HBs	negative	
HBsAg	positive	Chronically infected
anti-HBc	positive	
IgM anti-HBc	negative	
anti-HBs	negative	



Hepatitis C:

It is an Enveloped, single stranded-RNA flavivirus.

Mode of transmission is parenteral.

More than 50% of cases develop **chronic hepatitis**, there is increased risk of hepatocellular carcinoma.

Incubation period varies from 1-6 month.

Most common cause of transfusion associated hepatitis; sexual transmission is low.

For early diagnosis, Hepatitis C RNA by PCR

PCR (positive 1-2 weeks post infection) both quantitative and qualitative are available.

Quantitative PCR has a higher value for monitoring of therapy.

Hepatitis C Antibody (HCV-Ab):

HCV-Ab test is usually positive within 2-5 months after infection.

Interpretation of HCV-Ab tests

A positive HCV-Ab does not necessarily mean infection with HCV.

False positive results could be present due to for example: cross reactivity by an infection other than HCV.

Thus, Positive HCV Ab test should be confirmed by PCR.

Negative HCV Ab test does not exclude HCV 100%. False negative results are seen in those with immunodeficiency.

Also very early infection before seroconversion, HCV- Ab is negative, and you have to do PCR in this cases

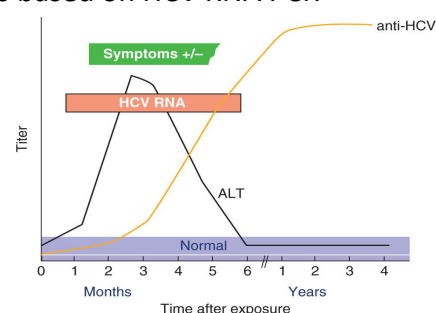
HCV-Ab (IgM)

Unlike HAV , HBV or other viral hepatitis in which IgM antibody is diagnostic of acute infection and precedes the appearance of IgG, for HCV infection the IgM responses are variably detected in both acute and chronic phases.

The IgM antibody response to HCV rarely precedes the appearance of IgG.

The IgM response may be absent ,late or persistent after HCV infection. Anti HCV IgM can not therefore serve as a diagnostic marker of acute HCV infection.

Diagnosis of very recent or acute HCV infection is based on HCV RNA PCR



Hepatitis D (Delta virus):

It is RNA defective or incomplete virus, It cannot replicate without the presence of HBV.

Mode of transmission is parenteral or sexual.

Chronicity and hepatocellular carcinoma could occur.

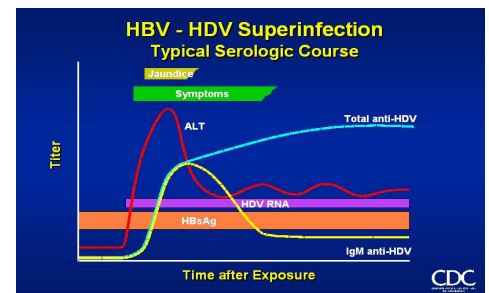
Cause infection only when it is encapsulated with HBsAg.

Cause 2 types of infections:

- 1- Acute co-infection after exposure to both HDV and HBV
- 2- Super infection of chronic carrier of HBV with a new HDV.

Incubation period varies from 4-8 weeks.

HDV is diagnosed by detection of HDV IgM antibody or by PCR.



Hepatitis E:

It is an Unenveloped single stranded RNA virus.

There are 4 genotypes of hepatitis E virus (HEV).

All can cause acute viral hepatitis.

Genotypes 1 and 2 usually cause waterborne outbreaks that are linked to fecal contamination of the water supply and fecal-oral person-to-person transmission.

Genotypes 3 and 4 typically cause sporadic cases rather than outbreaks.

Mode of transmission is fecal/oral and can involve eating uncooked or undercooked meat (as pork).

It never causes chronicity, It has a high mortality rate among pregnant women reaching 20%.

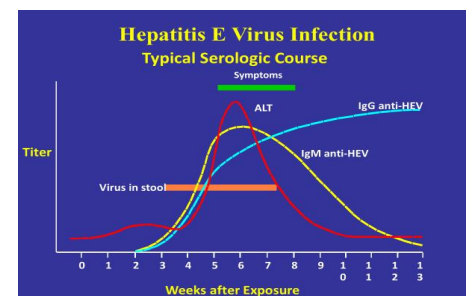
Like HAV it is usually self limited and does not cause chronicity. However, few reports document chronic genotype 3 hepatitis E exclusively in immunocompromised patients (including organ-transplant recipients, patients receiving cancer chemotherapy, and HIV-infected patients).

Incubation period varies from 2-8 weeks.

Cause enterically transmitted water borne infection

Primarily in adults,, rare in children

Diagnosis of recent infection is by detection of HEV-IgM antibody or by PCR.



Hepatitis G:

It is a single stranded RNA virus.

Mode of transmission is parenteral by contaminated blood or blood products.

Its prevalence in blood donors is rare (1%).

Most cases recovered completely, and chronicity is very rare or absent

Laboratory findings of liver cirrhosis:

- Albumin is low and gamma globulin is high with reversed A/G ratio, Total protein is variable
- Mild to moderate elevation of AST and ALT (AST>ALT).
- ALP is normal or mildly elevated.
- GGT is mildly elevated (especially alcoholic).
- Bilirubin is usually normal or elevated in late stages.
- PT is prolonged and not corrected by Vit K
- Decreased platelets count (due to hypersplenism and or decreased Thrombopoietin production from the liver). The latter is a glycoprotein hormone produced by the liver and kidney, which stimulates platelets production from megakaryocytes in the bone marrow.
- Serum protein electrophoresis (SPE): polyclonal increase of gamma globulin with beta gamma bridge (due to increase of IgA mainly). However, nowadays SPE is not commonly used for diagnosis of liver cirrhosis.
- Mild Increases in α -Fetoprotein (AFP) concentrations are common in cirrhotic and chronic hepatitis patients, even in the absence of hepatocellular carcinoma (HCC).

AFP is the most common tumor marker used for screening of HCC and is usually present at higher concentrations in HCC.

Hepatic infiltrations

- Due to space occupying lesions as in tumors, primary or secondaries, cysts, lymph nodes and others.
- Marked elevation of GGT and ALP.
- Bilirubin may be normal (partial obstruction allows passage of bilirubin and stimulates synthesis of ALP and GGT).
- Normal or slight increase of AST and ALT (AST>ALT)
- Tumor markers: $\uparrow$ AFP in HCC, $\uparrow$ CEA in GIT cancer with liver secondaries

Chronic active hepatitis

Definition: chronic inflammation of liver cells for at least 6 months.

It is commonly due to viral infection (HBV or HCV or HBV +HDV) or autoimmune or drug induced.

Clinical and laboratory features of chronic hepatitis are also observed occasionally in patients with such hereditary/metabolic disorders as Wilson disease (copper overload).

Laboratory findings of chronic hepatitis

- Albumin is **low** and **gamma globulin is high (IgG mainly)**.
- Mild to moderate elevation of ALT and AST (**ALT>AST**).
- Positive HBsAg, IgG anti-HBc and HBV DNA in chronic hepatitis caused by HBV. In some cases, HBeAg may be also positive.
- Positive HCV-Ab and HCV RNA in chronic hepatitis caused by HCV
- Auto antibodies may be found particularly in autoimmune type (ANA, ASMA, LKM). In chronic hepatitis due to viral infection, autoantibodies are uncommon or found at low titer
- Antinuclear antibody (ANA) and Anti smooth muscle antibody (ASMA) are commonly found in type 1 autoimmune hepatitis (often seen in adult particularly young women, It is the most common type)
- Anti-liver kidney microsome antibody type 1 (LKM 1) and anti-liver cytosol (anti-liver cytosol 1) are commonly found in type 2 autoimmune hepatitis (often affects pediatrics from 2-18 years. It is less common than type 1)

Differentiation between most common types of chronic hepatitis

Findings	Autoimmune	HBV	HCV
Sex	Female	Male	Equal
Age	15-25	Older	All
HBsAg	-ve	+ve	-ve
HCV-Ab	-ve	-ve	+ve
Autoimmune diseases	frequent	rare	occasional
Autoantibodies (ANA,ASMA, LKM)	High titre	Low or absent	Low or absent
Risk of cancer	Low	High	High

Serum Protein Electrophoresis--Normal Pattern

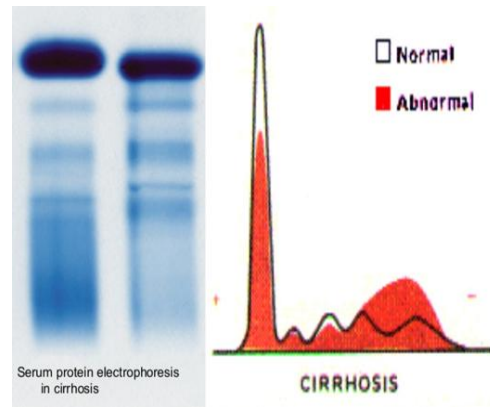
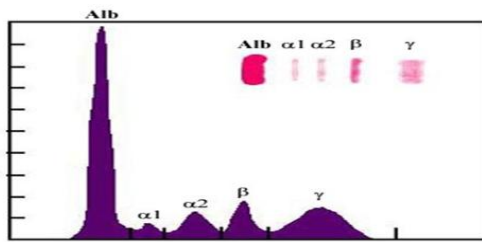


TABLE 10.2 Autoimmune Hepatitis Subtypes

Feature	Type 1	Type 2
Approximate frequency	95%	5%
Autoantibodies ⁹	ANA alone, 10% ASMA alone, 35% Both positive, 50% Both negative, 5%	Anti-liver/kidney microsomal antibody (LKM type 1) Anti-liver cytosol (LC-1)
Elevated serum IgG levels	+++	+
Most common affected age group	Children to adults (age 10 years to elderly)	Pediatrics (2–18 years)
Progression to cirrhosis (approximate %)	45	80
Histologic findings	Hepatitis with prominent plasma cells	Hepatitis with prominent plasma cells

ANA, antinuclear antibody; ASMA, antismooth muscle antibody.

QUIZ exam

1- Which of the following is associated with conjugated hyperbilirubinemia?

- A. Gilbert Syndrome.
- B. Crigler–Najjar syndrome.
- C. Dubin–Johnson syndrome.
- D. Hemolytic anemia.

2- Which of the following is a cause of unconjugated hyperbilirubinemia?

- A. Gilbert Syndrome.
- B. Dubin–Johnson syndrome.
- C. Cancer head of pancreas.
- D. Primary biliary cirrhosis.

Acute hepatitis

Acute damage of the liver cells can be caused by several agents:

- Viral infections : Virus A, B , D , C , E , Epstein- Barr virus , Cytomegalovirus.
- Non viral infection : Toxoplasmosis – Leptospirosis – Q fever.
- Drugs : Paracetamol.
- Alcohol.
- Poisons : e.g. Amanita phalloides (mushrooms).
- Other : Pregnancy , circulatory insufficiency , Wilson's disease.

Acute Viral Hepatitis

Acute viral hepatitis is the most common cause of liver disease in the world.

With its sequelae, Viral hepatitis is responsible for 1-2 million deaths annually.

Etiology:

- Enterically transmitted viruses: HAV, HEV.
- Blood-borne viruses: HBV, HDV and HCV.

Hepatitis A virus (HAV)

Hepatitis A is the most common viral hepatitis occurring worldwide, often in epidemics.

The disease is commonly seen in the autumn and affects children and young adults.

Spread of infection is mainly by the feco-oral route and arises from the ingestion of contaminated food.

HAV is a picornavirus, It replicates in the liver, and is excreted in bile and feces for about 2 weeks before the onset of clinical illness and for up to 7 days after.

The disease is maximally infectious just before the onset of jaundice.

Investigations:

Liver biochemistry:

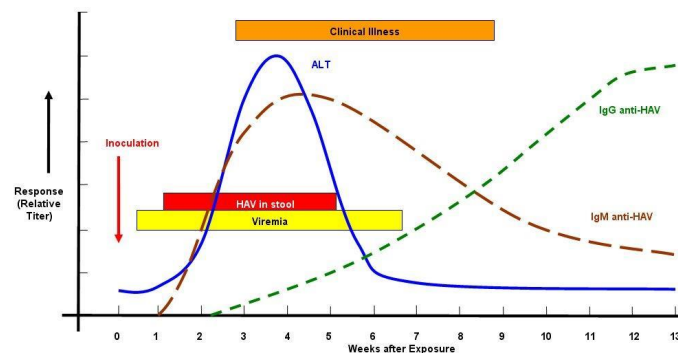
- In the prodromal stage: the serum bilirubin is usually normal, however, there is bilirubinuria and increased urinary urobilinogen. A raised serum AST or ALT, which can sometimes be very high, precedes the jaundice.
- In the Icteric stage: the serum bilirubin reflects the level of jaundice.

Haematological tests:

- There is leucopenia with a relative lymphocytosis.
- The prothrombin time (PT) is prolonged in severe cases. The erythrocyte sedimentation rate (ESR) is raised.

Viral markers:

An anti-HAV IgM means an acute infection, HAV IgG antibodies are common in the general population over the age of 50 years and has no role in diagnosis of acute infection, also in areas of high prevalence most children have antibodies following asymptomatic infection.



Time course of the events and serological changes of acute Hepatitis A infection

Other tests: Further tests are not necessary in the presence of an IgM antibody, but liver biochemistry must be followed to establish a return to normal levels.

Differential diagnosis:

Is from all other causes of jaundice, but in particular from other types of viral and drug-induced hepatitis.

Course and prognosis:

The prognosis is excellent, with most patients making a complete recovery.

The mortality in young adults is 0.1 % but it increases with age.

Death is due to Fulminant hepatic failure

During convalescence, 5-15% of patients may have relapse of the hepatitis but this settles spontaneously. Occasionally, a more severe jaundice with cholestasis will run a prolonged course and is called 'cholestatic viral hepatitis'.

Treatment is by reassurance. HAV hepatitis never progresses to chronic liver disease.

Treatment:

There is no specific treatment, and rest and dietary measures are unhelpful. Corticosteroids have no benefit. Admission

Prevention:

Hygiene (e.g., hand washing) and sanitation (e.g., clean water sources) Boiling, chlorination of water

Vaccination:

Active immunization:

HAV vaccine (inactivated vaccine) , it is effective& safe, the protection starts after 15 days and lasts for more than 20 years.

It is given in two doses followed by a booster at 6 or 12 months.

It is recommended for travelers to endemic areas, patients with CLD and close contacts of a patient.

Post-exposure: Theoretically immunoglobulins can be given to contacts of patients with acute HAV infections

Hepatitis E virus (HEV)

HEV is an RNA virus causing hepatitis clinically very similar to hepatitis A.

It is enterically transmitted, usually by contaminated water.

Epidemics have been seen in many developing countries and sporadically in developed countries, in patients who have had contacts with farm animals or travel abroad.

It has mortality from fulminant hepatic failure of 1-2%, which rises to 20% in pregnant women especially in the second and third trimester.

There is no carrier state and it does not progress to chronic liver disease except in some immunosuppressed patients in solid -organ transplant recipients.

An ELISA for IgG and IgM anti-HEV is available for diagnosis.

HEV RNA can be detected in the serum or stools by PCR.

Prevention and control depend on good sanitation and hygiene; a vaccine has been developed.

Hepatitis B Virus (HBV)

- It is a DNA, hepadna virus, there are 8 genotypes (A to H).
- It has an outer envelope: HBsAg.
- The HBV core contains: -HB core antigen (HBcAg) -HBe antigen (HBeAg)
- Partially double-stranded DNA -DNA polymerase/reverse transcriptase enzyme.

The transmission is by the parenteral, sexual and vertical route (at time of birth & close contact afterwards). Interfamily spread can also occur by household articles.

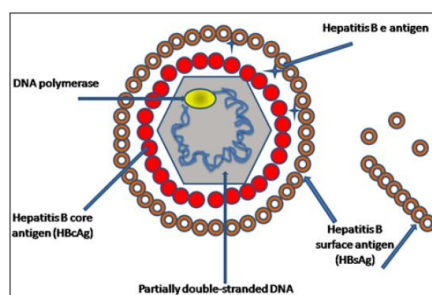
Incubation period: 2-6 months.

The clinical attack may be mild or severe.

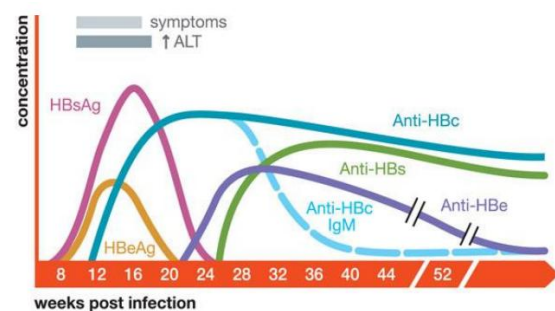
Serological diagnosis in the acute stage depends on IgM anti- HBV , HBVDNA and HBsAg.

After the attack, persistent infection may occur and can result in chronicity and cirrhosis and hepatocellular carcinoma.

Prophylaxis: hepatitis B immunoglobulin and vaccination.



Hepatitis B Virus



Time course of the events and serological changes of Hepatitis B infection

Hepatitis D virus (HDV)

It is an incomplete RNA virus, it is unable to replicate on its own but is activated in the presence of HBV.

Mode of transmission: blood- born, sexual, vertical.

Incubation period: 4-7 weeks. HDV infection is declining due to HBV vaccination.

It is endemic in Mediterranean basin.

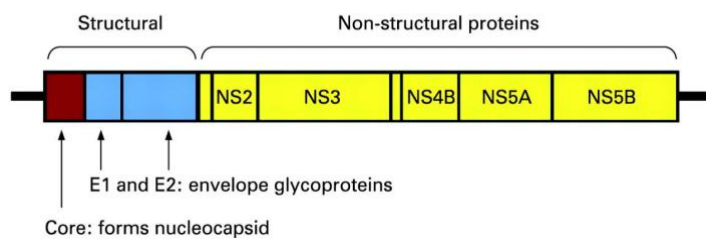
Infection occurs as:

- 1- Co-infection with HBV at the same time.
- 2- Super infection of a chronic HBV carrier. Synthesis of HDV depresses the appearance of HBV viral markers.

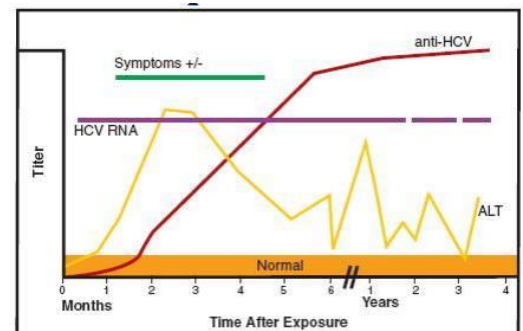
Prophylaxis: vaccination against hepatitis B.

Hepatitis C virus (HCV)

- HCV is an important cause of chronic liver disease worldwide, it was highly prevalent in Egypt.
- HCV is a single-stranded RNA of the flaviviridae family, the RNA genome is encoding a polyprotein product consisting of structural (capsid and envelope) and non structural viral protein.
- HCV has at least 8 major genotypes. Genotype 4 is the most prevalent in Egypt .



HCV genome



Serology and Virology of Acute and Chronic HCV

Incubation period: 1-6 months.

Transmission: mainly parenteral. There is a low frequency of sexual and vertical transmission

Age: usually above 16 years.

Clinical attack: Usually mild and sub-clinical .

The elevation of transaminases and the appearance of HCV RNA precede the appearance of HCV antibodies

Persistent infection is common (about 80%). Infection is linked with chronic hepatitis, cirrhosis and HCC.

Prophylaxis: Prevention of nosocomial infection and supervision of blood donation

General clinical Picture of viral hepatitis

1- Anicteric Hepatitis: Mild attack with no jaundice, it may presents as gastroenteritis and is usually under-diagnosed.

2- Icteric hepatitis:

Pre-icteric stage: It may lasts for few days.

Patient feels generally unwell with marked malaise, Anorexia & nausea Pain may occurs in right hypochondrium associated with fever.

This is followed by darkening of urine & light stool.

In hepatitis B serum sickness-like syndrome occurs in less than 10% of cases with manifestations of fever, maculopapular rash and polyarthrits of the small joints.

Icteric stage:

2-4 weeks, with the appearance of jaundice, symptoms decrease:

Fever decreases, appetite improves and vomiting stops.

Persistent vomiting and/or drowsiness or confusion indicates urgent hospital a as they may reflect liver failure.

Liver is enlarged in 70% of patients, soft & tender, splenomegaly and lymphadenopathy may develop in some patients .

Convalescence stage: Recovery is rapid in children. Fatigue persists for some weeks.

Clinical & biochemical recovery is usual **within 6 months**.

Sequelae of an acute attack of viral hepatitis:

1- Complete recovery is the rule in most HAV& HEV infections, also in more than 90% of HBV infections in adults . Only 20% of HCV patients recover spontaneously.

2- Relapse: Especially in HAV infections. The attack is usually milder attack, and is marked by elevation of transaminases and serum bilirubin. Recovery usually follows.

3- Prolonged cholestasis: commonly in HAV infection, the jaundice persists for several months. Pruritus is prominent. Abdominal U/S is usually performed to exclude surgical obstructive jaundice. Complete recovery is the rule.

4- Acute liver failure: Jaundice is deep, drowsiness, confusion , flapping tremors develop and coagulopathy.

5-Persistent infection:

- HBV carrier rate depends on age (see later).
- HDV super-infection causes chronic hepatitis in about 70%.
- HCV infections persist in about 80%.

Some of the characteristics of hepatitis viruses

	HAV	HBV	HCV	HDV	HEV
Genome	RNA	DNA	RNA	RNA	RNA
Incubation period	2-6 W	2-6 M	1-6 M	4-7 W	2-6 W
Serologic diagnosis	IgM anti HAV	- IgM anti HBV - HBVDNA - HBsAg	- Anti HCV - HCVRNA	IgM anti HDV	IgM anti HEV
Prevention	Vaccine	Vaccine	-----	Vaccine against HBV	-----
Chronicity	-----	+	+	+	It does not progress to chronic liver disease except in some immunosuppressed patients in solid-organ transplant recipients
Treatment	Non specific	Non specific + antiviral	Non specific + antiviral	- Pegylated interferon - Bulevirtide	Non specific

Pathology

- All forms of acute viral hepatitis have a basic pathology.
- Acute inflammation of the entire liver
- Centrizonal necrosis & cholestasis are most marked in zone 3.
- Cellular infiltration (mainly lymphocytes) of portal tracts (zone 1) and bile duct proliferation.
- The reticulin framework is usually preserved & liver cells regenerate from zone 1.
- **Other organs:** There may be regional lymph nodes enlargement, Splenomegaly and hypoplasia in bone marrow

Investigations

1- Liver function tests:

- Marked elevation of ALT, AST (500-5000U/L).
- **Serum bilirubin** is increased (both direct and indirect).
- Alkaline phosphatase and Prothrombin time are normal or mildly increased.
- Albumin is normal, **γ globulins are increased**.

2- Blood changes:

- Leucopenia with relative lymphocytosis.
- ESR is increased.

3- Urine and stool:

- Bilirubin appears early (dark urine).
- Urobilinogen is variable.
- The Stools appears pale due to decreased stercobilinogen.

4- Viral markers:

- HAV:

HAV IgM: indicates recent infection and is positive for 3-6 months thereafter.

HAV IgG: indicates old infection and immunity.

- HBV:

HBsAg: appears 6 weeks after infection & disappear in 3 months. Persistence for more than six months indicates the development of a carrier state.

Anti-HBs: appear after 3months. It means recovery & immunity.

HBc IgM: is present in acute phase & in severe chronic hepatitis.

HBc IgG: Past exposure.

HBeAg: viral replication and infectivity.

Anti-HBe: low infectivity.

HBV-DNA: most sensitive marker of viral replication, measured by PCR.

- HDV:

HDV IgM and **HDV RNA** in acute infections.

HDV IgG in chronic phase.

- HCV:

HCV antibodies become positive after 6-8 weeks of infection.

HCVRNA can be detected 1-2 weeks after infection by PCR.

- HEV:

HEV IgM indicates recent infection.

HEV IgG old infection.

Treatment (acute attack)

Rest: Bed rest is mandatory until the patient is free of jaundice & free of symptoms and the bilirubin is less than 1.5mg/dl.

Diet: traditionally the patient was given a low fat, high CHO diet as it is said to be the most appetizing.

Prevention

Pre-exposure:

HAV vaccine (inactivated vaccine) ,(previously discussed).

HEV: Efficiency of immunoglobulins is uncertain. An effective vaccine has been developed.

HBV: Pre-exposure HBV recombinant vaccines (or Plasma derived) are both effective & safe.

- They contain HBsAg as the immunogen.
- For the recombinant vaccine: the dose is 1 ml IM in the arm at 0-1-6 months.
- Indications:
 - ◆ Surgical, dental, laboratory staff & medical students.
 - ◆ Patients: dialysis, oncology, CLD, mental patients.
 - ◆ Neonates of HBsAg mothers.
 - ◆ Accidental exposure.
 - ◆ Drug users & homosexuals.

Post-exposure:

Both HBV vaccine and HB immunoglobulin (HBIG) are given in different arms at the same time to: neonates of HBsAg positive mothers, accidental exposure, HBsAg +ve blood, to sexual contacts of acute sufferers and to prevent reinfection of a donor liver inserted into an HBV DNA positive patient.

HCV : Work on HCV vaccine is in progress, Screening of blood donors and blood banks for HCV antibodies.

Herpes virus hepatitis

- A large family of enveloped DNA viruses including cytomegalovirus (CMV), Epstein-Barr virus (EBV), herpes simplex virus (HSV).
- Very common infections ,90 to 95% of adults are (+) for EBV, and CMV serology, and up to 80% for HSV
- Unlike hepatitis viruses A, B, C, D and E, herpes viruses are not hepatotropic, and do not replicate in liver cells but can cause hepatitis by proliferation of infected cells and release of cytokines.
- They do not cause chronic hepatitis or cirrhosis.
- EBV and CMV hepatitis are very common and usually mild, subclinical and self-limited.
- Herpes-virus hepatitis can present as a primary infection or reactivation of latent infection

EBV and CMV

Transmission: Person to person transmission, uses of contaminated objects, Blood transfusion, and organ transplantation have also been reported as rare modes of transmission.

EBV symptoms

Non-specific: malaise, anorexia, right sided abdominal pain

characteristic: Infectious mononucleosis (IM) syndrome

- sore throat with exudative pharyngitis, enlarged lymph nodes.
- EBV is especially associated with left sided abdominal pain with splenomegaly.

Diagnosis by Serology:

- viral capsid antibody (VCA IgM)
- Heterophil Ab (Specific but not very sensitive) EBV DNA by PCR .
- Lymphocytosis, atypical lymphocytes (not specific)

CMV Symptoms

Similar to EBV (infectious mononucleosis), sore throat, enlarged lymph nodes.

Exudative pharyngitis and splenomegaly are less common than with EBV.

Diagnosis

by **Serology** , IgM Ab to envelope (+) ,Heterophil (-) ,CMV DNA by PCR ,Lymphocytosis, atypical lymphocytes – not specific

EBV and CMV biochemistry:

- Usually only mildly elevated ALT and usually ALT >AST.
- Elevated alkaline phosphatase in 60%, hyperbilirubinemia in 45%, but jaundice in usually <5% of cases
- The laboratory abnormalities typically normalize in a few weeks, but can rise again during reactivation.

Prognosis

Severe cholestatic hepatitis, granulomatous hepatitis, and rarely, acute fulminant hepatic failure - possible especially in immunocompromised.

Treatment

Most cases of EBV, and CMV hepatitis are mild, resolve spontaneously - only supportive treatment is needed. In cases of severe hepatitis, valganciclovir or ganciclovir can be used

HSV Hepatitis

Rare, usually a result of disseminated herpes sepsis.

High risk patients:

Recipients of organ transplants on immunosuppressive therapy, chemotherapy and persons with immunodeficiency diseases like – high risk.

Pregnancy, 2nd or 3rd trimester

- Common cause of **Fulminant hepatic failure** even in immunocompetent patients.

Symptoms

Herpetic cutaneous, muco-cutaneous, mucosal lesions (cold sores) patient presents with severe hepatitis

AST, ALT – very high, >10 X ULN .

Fulminant hepatic failure frequent especially, but not only, in immunocompromised

Treatment

Because of potential rapid deterioration and high mortality of HSV hepatitis ,empiric treatment with acyclovir or Foscarnet is recommended as soon as possible after diagnosis to decrease mortality and the need for liver transplant.

Chronic hepatitis

This is a group of liver disease characterized by diffuse inflammation with necrosis lasting for more than six months .Inflammation of the liver may progress from acute inflammation to chronic inflammation and finally leads to cirrhosis.

- Whatever the etiology, the histology is nearly the same.

Etiology

- Viruses: HCV, HBV $\pm$ HDV.
- Autoimmune hepatitis.
- Drugs: α methyl dopa , isoniazide , methotrexate , nitrofurantoin.
- Alcoholic steatohepatitis.
- Metabolic : Non Alcoholic steatohepatitis (NASH) , rarely Wilson's disease & α 1 antitrypsin deficiency and hemochromatosis.

Pathology

In the mild form: The infiltration by the inflammatory cells: lymphocytes & plasma cells could be seen in the portal area , erosion of the limiting plate, piecemeal necrosis of the periphery of the lobule (interface hepatitis) or focal necrosis could be seen inside the lobule.

In the severe form, the fibrous septa extend inside the lobule to form rosette formation and bridging fibrosis. Cirrhosis may be present.

The Metavir Score helps interpretation of the liver biopsy.

This scoring system assigns two standardized letters: one to represent the degree of inflammation (A) and the other the degree of fibrosis (F).

The Activity score scales from A0 (no activity) to A3 (severe activity) and the Fibrosis score scales from F0 (no fibrosis) to F4 (cirrhosis).

Chronic hepatitis B

This is common in developing countries. Its overall prevalence in Egypt is 5%, mainly genotype D .

Clinical picture

- More common in males.
- Chronicity depends on age and immune status of the patients. Following an acute HBV infection, which may be subclinical, approximately 1-10% of patients will not clear the virus and will develop a chronic HBV . This occurs more readily with neonates (90%) or children (20-50% below the age of 5 years) than when HBV is acquired in adult life (<10%).
- Symptoms: range from asymptomatic, fatigue, abdominal pain to advanced liver cirrhosis with its associated complications.
- About 50% of HBsAg carriers have evidence of viral replication (HBeAg+/HBV DNA+).
- About 20% of HBV carriers develop cirrhosis within 5 years of disease onset.
- Most damage to the liver is caused by the host immune response to the virus rather than by the virus directly.
- Three different clinical pictures which could be observed in chronic HBV infection :

Immune tolerant chronic HBV infection:

Patients are asymptomatic and HBsAg and HBeAg positive with very high levels of serum HBV DNA, but with normal liver function tests.

Most have been infected prenatally.

They do not require treatment, but often in the third to fifth decade in life there is a change in host immunity , so that acute hepatitis develops and chronic hepatitis can follow.

Inactive carriers with chronic HBV infection:

Usually discovered when they are screened before donating blood for transfusion.

Patients have HBsAg in their serum and are HBeAg negative, HBe antibody positive, with levels of HBV DNA, which are below 400 IU/L in their serum.

They have normal aminotransferase levels.

Most remain HBsAg positive, but do not develop progressive disease, although some patients may rarely have episodes of reactive hepatitis such as during chemotherapy for cancer or with bone marrow transplantation, so antiviral prophylaxis is indicated.

There is an annual spontaneous clearance rate of HBsAg of 1-2%.

Active chronic HBV infection:

These patients have raised serum aminotransferases with evidence of HBV replication; they have HBe antigen and/or anti-HBe and HBV DNA in their serum.

A liver biopsy shows chronic hepatitis.

Investigations

- Diagnosis of HBV infection depends on the detection of HBsAg.
- ALT and AST may be elevated.
- Liver biopsy: is indicated to determine the grade of fibrosis and level of activity. Hepatocytes with **groundglass** appearance represent the production of HBsAg in cytoplasm.

Treatment

Indications for therapy are similar for HBeAg positive or negative patients with chronic hepatitis.

Three criteria are used:

- serum HBV DNA levels
- serum ALT levels and histological grade and stage
- Patients with moderate to severe active necroinflammation and/or fibrosis in the liver biopsy with HBV DNA above 2000 IU/ml and/or ALT above the upper limit of normal.

Age and co-morbidities also affect the decision to treat and with which agents.

- **If cirrhosis is present**, treatment should be given independent of ALT or HBV DNA levels. Patients with decompensated cirrhosis can also be treated with oral antiviral agents
- **Immunotolerant patients**, usually young with persistently normal ALT and high HBV DNA levels, without evidence of liver disease, and without a family history of cirrhosis or hepatocellular cancer do not need therapy, but must be followed up regularly.

The aim of treatment is the seroconversion of HBeAg when present to anti-HBe, and the reduction of HBV DNA to the lowest possible level as measured by sensitive PCR techniques, In addition normalization of serum ALT, histological improvement of inflammation and fibrosis, and loss of HBsAg reflect a good response.

If HBeAg disappears, remission is usually sustained for many years.

Patients usually remain HBsAg positive, but there is a small loss of HBsAg over the years Interferon and antiviral agents such as **entecavir** and **tenofovir** are the most commonly used drugs.

Response to therapy is judged by a reduction in the HBV DNA level, and if HBeAg is present, by seroconversion to anti-HBe.

Pegylated α 2a interferon (180 μ g once a week subcutaneously) gives response rates of 25-45% after 48 weeks of treatment.

Younger patients with higher serum aminotransferase values (3x the upper limit of normal), with viral loads <10 IU/ml respond best.

Side-effects of treatment Influenzal like symptoms and bone marrow depression may also occur.

Overall, the response rate (i.e. disappearance of HbeAg) is 25-40%.

Oral therapy:

- Tenofovir 300 mg tablet /day or Tenofovir alafenamide (TAF) 25 mg tablet / day very effective, and with no resistance mutants have been described. It can also be used in patients with lamivudine mutations, but not on its own for adefovir mutations. Tenofovir is a very safe drug , however it is advised to monitor bone mineral density and kidney functions tests in long term use .
- Entecavir 0.5 – 1 mg /day is very effective and reduces HBV DNA quickly, and there is little viral resistance. Serum HBV DNA becomes negative in 67% (HBeAg-positive patients) and 90% (HBeAg-negative patients) by 48 weeks.
- Lamivudine, 100 mg/day given orally, is well tolerated. However there is the problem of viral resistance due to YMDD mutant .
- The duration of all treatment is probably lifelong and is still being assessed.
- Hepatitis Delta could be treated by Pegylated interferon alpha or by a new oral antiviral Bulevirtide

Chronic hepatitis C

Clinical features

- Most acute infections are asymptomatic .
- Patients are identified years later and usually accidentally (elevated liver enzymes or with positive HCV antibody)
- HCV Infection has the propensity to become a chronic hepatitis in 80%of cases .
- Patients may be: asymptomatic or present with fatigue , or with upper abdominal pain, they may also presents by manifestations of liver cirrhosis and its complications, There is a long period between infection and cirrhosis (usually 20 years) .
- Hepatocellular carcinoma occurs in about 15% of cirrhotic patients.
- Extrahepatic manifestations: may be due to deposition of immune complexes

The following clinical manifestations can occur:

- Mixed cryoglobulinemia
- Porphyria cutanea tarda
- Glomerulonephritis
- Vasculitis& others.

Diagnosis

- **Viral markers:** HCV Ab by ELISA to indicate infection or exposure.
- HCV RNA by **PCR** to indicate viral replication.
- **Liver biopsy** is variable: and may show minor pathological changes or show cirrhotic changes, Fatty changes and lymphoid follicles in the portal tracts are frequently present.
- ALT may be normal/ elevated or fluctuating.

Treatment

The strategy of treatment of HCV patients has been completely revolutionized by the discovery of the direct acting antiviral drugs (DAAs) , and now an attempt to treat most patients with HCV infection is recommended .

The available drugs for treatment of chronic HCV are :

I- The Direct-Acting Antivirals (DAAs)

II- Ribavirin (RBV)

III- Pegylated Interferon (IFN)

I- The Direct-Acting Antivirals (DAAs): DAAs target specific non-structural (NS) proteins that are vital to the replication of HCV and result in disruption of viral replication and infection. These include

- Sofosbuvir (Sovaldi): Highly potent against all genotypes , till now it represents the backbone of most treatment regimens , 400mg tablet once daily with or without food, No dose adjustment, but must be used with care for patient with advanced liver disease, Adverse effects in 10% of cases: fatigue, headache, anorexia & insomnia.
- Daclatasvir : 1 tablets, 60mg each, are given once daily
- Ledipasvir: Available only in fixed dose combination with Sofosbuvir

Fixed dose Combinations:

- Sofosbuvir-Ledipasvir (Harvoni): One tablet (400 mg Sofosbuvir-90 mg Ledipasvir), once daily.

II- Ribavirin (RBV): Used as adjuvant to other drugs. Dosed varies from 600 to 1200mg/day orally according to weight. AE: hemolytic anemia, teratogenesis & causes sperm abnormalities.

III- Pegylated Interferon (IFN): Pegylated interferon: α 2a (Pegasys) 180 μ g/week SC, or α 2b (Pegintron) 1.5 μ g/Kg/ week SC, Adverse effects: marked fatigue and body pains, loss of hair, pancytopenia and others.

The new treatment trend is for IFN free regimens, due to its severe AE.

Egypt and HCV

HCV was very prevalent in Egypt.

The prevailing genotype is the genotype 4.

This high prevalence rate is partially related to the nosocomial infection due to medical procedures performed without appropriate cleaning or disinfection of the used equipment such as in circumcision, dentistry practices , also due to the prevailing social and cultural behavior.

The high HCV prevalence has been linked to the use of contaminated and inadequately sterilized needles in the campaign of parenteral anti schistosomal therapy in the past

Egypt's national response to this high prevalence was manifested by the creation of the national Committee for control of viral hepatitis (2007) and by the creation of the Egyptian national control strategy for viral hepatitis and for trial for standardization of treatment protocols by putting national guidelines for HCV treatment.

The government sponsored centers (more than 32 affiliated units) and the health insurance treatment service represent the main access of treatment for the Egyptian patients.

The construction of a national network of treatment centers (NNTC) through the internet allowed easy registration for all patients all over the country to receive the anti HCV treatment, and allowed the collection of research data which served in the modification of the national guidelines.

More than 350 000 patients were treated by the old regimen pegylated IFN-Ribavirin

The price of the brand antiviral drugs starting from the era of pegylated IFN and RBV to the DAAs is lower in the Egyptian market in comparison to other countries in the world, and there is progressive reduction of drug prices .

During the period October 2018 – May 2019 , Egypt launched a large campaign for screening and treatment of HCV aiming for future eradication of the disease.

The screening also aimed for detection of non-communicable diseases among population (hypertension , DM , Obesity).

Screening was done for more than 49 million of the population , the prevalence of HCV was 4.4 % all over the country. Eligible patients were treated

There is an urgent need for strengthening prevention efforts, and bringing down the costs of antiviral drugs in Egypt where the most precarious situations are also those most likely to be infected by the virus

Current Egyptian program for treatment of HCV

Patients were categorized to

1- Easy to treat group: Treatment Naive S bilirubin ≤ 1.2 mg/dl- Serum Albumin ≥ 3.5 g/dl – INR ≤ 1.2 , Platelet count $\geq 150\,000/\text{mm}^3$.

2- Not easy to treat group: Peg – IFN experienced , - S bilirubin ≥ 1.5 mg/dl Serum Albumin ≤ 3.2 g/dl – INR ≥ 1.5 , Platelet count $\leq 100\,000/\text{mm}^3$.

- Easy to treat patients are eligible for treatment by dual therapy : Sofosbuvir + Daclatasvir for 12 weeks
- Not easy to treat patients are eligible for treatment by triple therapy with Sofosbuvir + Daclatasvir + Ribavirin for 12 weeks
- Patient with relapse after treatment with a sofosbuvir based regimen are retreated with sofosbuvir + Daclatasvir + Ribavirin for 24 weeks
- Patient with chronic Kidney disease are treated by the usual treatment regimen
- Other options for treatment of HCV – genotype 4

Sofosbuvir- Ledipasvir (Harvoni): Can replace the combination Sofosbuvir-Daclatasvir

- The patient is cured when HCV RNA is negative 3 months after the end of treatment and this is called sustained virological response (SVR),
- Patient cured from HCV were asked to be vaccinated against hepatitis B

Autoimmune hepatitis

This is a chronic liver disease of unknown cause.

It is characterized the presence of circulating autoantibodies, increased gammaglobulins and interface hepatitis.

Near 50% increase in incidence over the last few decades

- Occurs at all ages and within all ethnic groups around the world

Types of autoimmune hepatitis (based on serological markers):

Type 1: Characterized by detection of serum anti-nuclear antibody (ANA), anti-smooth muscle antibody(ASMA).

Type 2: Characterized by detection of serum anti- liver- kidney microsomal antibody(LKM-1), anti-liver cytosole antibody-1(LC I)

Clinical Picture and diagnosis

Autoimmune hepatitis usually occurs in females (70%) mostly between 15-25 years or at menopause.

- Broad range from asymptomatic to acute/severe disease
- Diagnosis of AIH relies particularly on the presence of autoantibodies, hypergammaglobulinaemia and typical or compatible histology
- Elevated IgG levels, especially in the absence of cirrhosis, are a distinctive feature of AIH
- AIH should be considered in any patient with acute or chronic liver disease particularly if hypergammaglobulinaemia is present and if the patient has features of other autoimmune diseases

Typical symptoms are fatigue, joint pains, hepatomegaly and jaundice.

Associated autoimmune manifestations: polyarthrititis, hemolytic anemia, purpura, DM, autoimmune thyroiditis.

- γ globulins are increased mostly IgG (>1.5 folds ULN).
- AST & ALT are elevated. Alkaline phosphatase is not markedly increased
- Serum auto antibodies: ANA, SMA, LKM-1, LC-1.
- Liver biopsy: interface hepatitis and plasma cell infiltration, no biliary lesions.
- Other causes of liver disease must be excluded : viral hepatitis, drug induced liver diseases (DILI) ,Wilson's disease, consumption of alcohol, ,NASH , alpha-1 antitrypsin deficiency, hepatotoxic drugs, PBC, PSC, genetic hemochromatosis must be excluded .

Treatment

Aims:

To obtain complete remission of the disease and to prevent its further progression, this requires complete normalization of aminotransferases and IgG levels .

Generally the treatment requires permanent maintenance therapy

Absolute treatment indications:

Clinical

Incapacitating symptoms

Progression of disease

Laboratory

AST $\geq 10 \times \text{ULN}$

AST $\geq 5 \times \text{ULN}$ + GG $\geq 2 \times \text{ULN}$

Histology

Bridging necrosis

Multilobular necrosis

1- Corticosteroids improve symptoms, signs, LFTs & histology.

Dose: prednisone 0.5-1 mg/ Kg Bwt or 20-40 mg / day for 2 weeks then reduce to a maintenance dose for 4 weeks in combination with azathioprine (1-2 mg /Kg Bwt or 50-150mg/day).

2- Azathioprine , cyclosporine and other immunosuppressive drugs have been tried alone or with steroids as steroid sparing agent.

Disease remission:

- Disappearance of symptoms
- Normalization or near normalization of AST to $< 2 \times \text{ULN}$
- GGT and bilirubin: normal
- 65% achieve remission within 18 months, and 80% within 3 yrs of initiation of Rx

Liver transplantation: is another option , however the condition may recur.

Prognosis

Good prognosis, 10 year survival: 90%

Fatty Liver and nonalcoholic steatohepatitis

- Nonalcoholic fatty liver disease (NAFLD) refers to a wide spectrum of liver diseases ranging from simple fatty liver (steatosis) to nonalcoholic steatohepatitis (NASH).
- Recently renaming was made for non-alcoholic fatty liver disease (NAFLD) to metabolic dysfunction-associated steatotic liver disease (MASLD.) and renaming was made for nonalcoholic steatohepatitis (NASH) to metabolic dysfunction associated steatohepatitis (MASH).

- **Steatosis**: accumulation of fat in the liver (mainly triglycerides).

Steatosis alone is, considered a benign condition, Hepatic steatosis is usually caused by hyperinsulinemia in the setting of insulin resistance.

- **Nonalcoholic steatohepatitis (NASH)**: inflammation and necrosis of liver cells may be caused by the toxicity of excess free fatty acids, oxidant stress, or mitochondrial dysfunction.

This may lead finally to fibrosis, cirrhosis and even hepatocellular carcinoma.

- **NAFLD**: is a common liver disease worldwide as it is associated with the epidemic obesity.

NAFLD is a part of the metabolic syndrome which includes: Obesity (central obesity) Diabetes mellitus type 2. Insulin resistance Hypertension , Hyperlipdemia.

Clinical picture

- Typically asymptomatic.
- Equal sexes
- Fatigue/ right upper quadrant pain
- Hepatomegaly

Diagnosis

- ALT , AST : normal or increased.
- U/S: bright liver.
- Check blood glucose& serum TG.
- Exclude viral, drug, autoimmune & other causes.

Fibroscan device

- Measures liver compliance non-invasively using sound waves.
- Can distinguish between advanced cirrhosis and no cirrhosis.
- Difficulty with distinguishing steatosis from cirrhosis.

Liver biopsy - gold standard – can distinguish between steatosis and NASH, and various stages of fibrosis, but is invasive.

Management

- Gradual& sustained weight reduction (not more than 1.6 Kg/week) by: increase vegetables& fruits/ decrease fat.
- Caution on rapid weight loss – increased fibrosis.
- Lifestyle modifications: Exercise and Diet therapy.
- Most studies indicate weight loss improves enzymes and/or histology.
- Bariatric surgery for morbidly obese patients (BMI>35-40).
- Use of insulin sensitizing agents such as metformin or pioglitazone.
- Medications: no medications have been proven to be effective, or approved for this indication , however these medications are under clinical trials : Obeticholic acid , Liraglutide ; Semaglutide , Empagliflozin
- Management of co-morbid factors: DM, hypertension, hyperlipidemia

Liver cirrhosis

It is a diffuse process characterized by necrosis of liver cells, followed by fibrosis, regeneration and nodule formation.

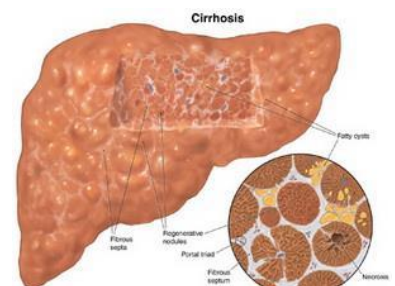
The liver architecture is diffusely abnormal and this interferes with liver blood flow and function.

This finally leads to impaired liver function.

Etiology

It is caused by chronic non self-limited injury of the liver:

- Viral hepatitis (chronic hepatitis C, Chronic hepatitis B±D).
- Non-alcoholic steatohepatitis (NASH).
- Alcohol: Prolonged (>10 years) and regular alcohol intake (>30 ml/day).
- Metabolic: Wilson's disease, hemochromatosis, α 1 antitrypsin deficiency,).
- Biliary cirrhosis: primary and secondary.
- Cardiac cirrhosis: hepatic venous outflow obstruction.
- Autoimmune hepatitis.
- Toxins & drugs: Methotrexate- Isoniazide- methyl dopa.



Morphological classification

1- Micronodular cirrhosis: small nodules, nearly same size (<3mm) usually caused by ongoing alcohol damage.

2- Macronodular cirrhosis: big nodules of variable size (>3mm), usually follow chronic hepatitis.

3- Mixed type

Clinical picture

1- Compensated cirrhosis:

The disease is discovered accidentally on routine examination and by screening.

Examination may reveal an enlarged firm liver, splenomegaly, palmer erythema, spiders.

Liver function tests are nearly normal.

2- Decompensated cirrhosis:

There are several clinical manifestations of cirrhosis that include liver cell failure: ascites, jaundice, encephalopathy...etc.

Symptoms

There are usually three modes of presentation:

1- Asymptomatic: The patient may remain non symptomatizing for years to be discovered by chance during routine check-up, pre-employment or pre-operative assessment

2- Symptoms of conditions commonly associated with chronic liver disease such as gall stones, recurrent pancreatitis (particularly in alcoholics), parotid enlargement, Dupuytren's contractures and muscle cramps

3- Symptoms of complications e.g. ascites, bleeding, encephalopathy, recurrent lower limb edema, spontaneous bacterial peritonitis or renal impairment.

4- Fever due to infections: UTI, SBP, chest and skin infections , or due to portal bacteremia and necrosis of the liver cells.

Signs

1- Protein-calorie malnutrition: found in 20-40%. of patients as shown by muscle wasting and decreased fat stores.

2- Skin: Spider naevi and Palmer erythema may be detected.

3- Nail changes:

a. Disappearance of nail lacunae

b. White opaque nails (leuconychia)

c. Clubbing of fingers is more commonly found in biliary cirrhosis and in cirrhosis associated with large pulmonary arterio-venous shunt.

4- Manifestations of hyperdynamic circulation due to the presence of excess vasodilator materials (VDM) leading to warm hands, high volume pulse, tachycardia.

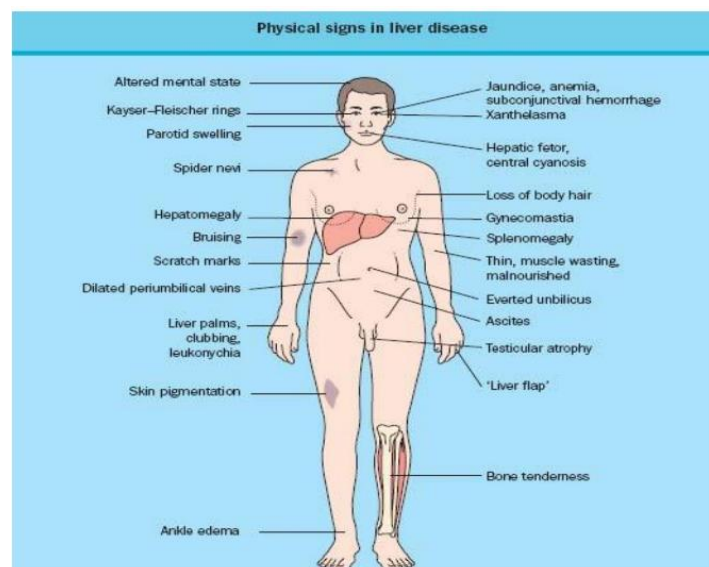
5- Manifestations of hormonal disturbance: feminization in male observed in long standing cases. This usually indicates decompensation.

Abdominal examination:

- 1- Abdominal wall hernias are common particularly umbilical hernia.
- 2- Liver is usually palpable with a sharp edge and firm consistency. The surface is usually smooth except in cases of macronodular cirrhosis or carcinoma. The liver size may become shrunken in late cases.
- 3- Spleen is usually palpable, it varies in size, its lower end may reach to iliac fossa in cases of hepatosplenic schistosomiasis.
- 4- Ascites may be detected.

Chest examination

- Pleural effusion in association with cirrhosis and in the absence of primary pulmonary or cardiac disease. It is usually right sided (70%) with ascites but sometimes without ascites.
- The hepatopulmonary syndrome may be present
- Central cyanosis, clubbing, and dyspnea due to:
 - pulmonary AV shunts
 - porto-pulmonary shunts
 - basal lung collapse due to tense ascites.
- Fetor hepaticus: a fecal smell due to lack of detoxification of mercaptan absorbed from colon.



Physical signs in liver cirrhosis

Investigations

A- Laboratory investigations

1- Blood picture:

- Microcytic hypochromic anemia is common in patients having gastrointestinal hemorrhage.
- Normochromic macrocytic anemia may be found.
- Leucopenia and thrombocytopenia in patients with hypersplenism.
- Prolonged prothrombin time that is not corrected by vitamin K injections indicating failure of the liver to synthesize prothrombin complex.

2- Serum albumin is decreased, serum gamma globulins are increased, and bilirubin is increased.

3- Serum aminotransferases (ALT and AST) are mildly elevated.

4-Blood glucose: impaired glucose tolerance may be found in 50% of patients and frank diabetes in 15-30% of patients, particularly in post-hepatitis C cirrhosis.

5- Increased **urobilinogen** in urine.

B- Abdominal ultrasound

The liver may show a coarse echogenic pattern.

The edge may be irregular and in advanced cases, the liver may be shrunken and portal hypertension may be shown by the dilated portal vein.

Splenomegaly and ascites may be detected.

C- Upper endoscopy for detection of varices

D- Investigations for the cause:

Viral markers, serum autoantibodies, serum iron-and ferritin, copper and ceruloplasmin, liver biopsy could be done to reach the possible cause of cirrhosis.

E- Non-invasive tests for the diagnosis of hepatic fibrosis, cirrhosis and steatosis:

Non-invasive tests has now replaced the role of liver biopsy in the diagnosis of fibrosis.

These approaches can overcome the limitations of liver biopsy and have become more widely used in routine clinical practice.

Imaging-based elastography and serum biomarkers are currently the two main non-invasive tests for liver fibrosis staging.

Liver stiffness measurement by imaging techniques such as transient elastography (Fibroscan).

Simple biomarker blood tests for fibrosis, including the aspartate aminotransferase-platelet ratio index (APRI) and fibrosis-4 index (FIB-4) score.

Treatment of cirrhosis

Actually the condition of management of hepatic fibrosis and portal hypertension has changed with the development of 2 new definitions

- Compensated advanced chronic liver disease (cACLD).
- Clinically significant portal Hypertension (CSPH).

Transition from the compensated to the decompensated stage in chronic liver disease is clinically marked by the development of complications such as ascites, variceal hemorrhage and overt hepatic encephalopathy.

Among patients with cACLD, at least two different stages have been identified based on the presence or absence of clinically significant portal hypertension (CSPH).

Patients with cACLD who develop CSPH are at high risk for hepatic decompensation and mortality if left untreated.

The use of elastography in clinical practice has enabled the early identification of patients who are at risk of having CSPH and decompensation.

Patients with Liver stiffness measurement ≤ 20 kPa and a platelet count $> 150 \times 10^9/L$ do not need upper tract gastrointestinal endoscopy, because the likelihood of large varices is low ($< 5\%$).

Also patients with a liver stiffness measurement value by transient elastography of > 25 kPa is sufficient to diagnose CSPH and also do not need to do upper gastrointestinal endoscopy and are treated with carvedilol to decrease the risk of cirrhosis decompensation.

Although the development of cirrhosis is an irreversible process, many lines of treatment can be provided to the patient ensuring a better quality of life and may promote regeneration of normally functioning hepatocytes.

• Nutrition:

- Calories: caloric requirements are supplied as normal individuals.
- Good intake of carbohydrates and moderate amounts of fat should be supplied.
- Proteins: are provided in an amount of 1-1.2 g/kg, to be increased to 1.5 g/kg in poorly nourished patients.
- In patients with encephalopathy, proteins should be restricted to 30-40 gm/day and are then increased when his consciousness improves.
- Salt restriction in ascitic patients.

• Treatment of underlying cause:

- Antiviral drugs in early cases with active inflammation and Corticosteroids in autoimmune hepatitis.
- Abstinence of alcohol intake.
- Chelation of copper and iron in Wilson's disease and hemochromatosis respectively.

• Treatment of complications: ascites, encephalopathy, bleeding, spontaneous bacterial peritonitis and renal impairment.

- Liver transplantation is the treatment of choice if available.

Course and Prognosis

This is extremely variable and depends on several factors such as the etiology, and the presence or absence of complications.

The Child-Turcotte Pugh (CTP) Score is used to assess the prognosis, It is proposed to assess survival, risk of general anesthesia, and operative mortality.

Patients are classified into A, B & C according to 5 parameters: The level of albumin, bilirubin, prothrombin time & the presence of ascites& encephalopathy

Parameter	1 point	2 points	3 points
Encephalopathy	None	Grade 1-2	Grade 3-4
Ascites	None	Medically controlled	Uncontrolled
Albumin	>3.5	2.8-3.5 g/dl	< 2.8 g/dl
Bilirubin	<2	2-3 mg/dl	>3mg/dl
International normalized ratio (INR)	< 1.7	1.7-2.3	> 2.3

A patient is considered to be :

Child class A if he has < 7 points.

Child class B if he has 7-9 points.

Child class C if he has > 10 points.

The Model for End-Stage Liver Disease (MELD): It is a prognostic assessment based on the serum bilirubin, serum creatinine& INR. It is used to determine optimal timing for liver transplantation.

Primary biliary cirrhosis (PBC)

Definition

This is a condition in which there is progressive destruction of bile ducts leading to chronic cholestasis and eventually cirrhosis.

The cause is not yet identified but it is said to be an immunological disease also there is a familial background

It is sometimes associated with autoimmune manifestations in the form of rheumatoid arthritis, and Sjogren's syndrome.

Pathogenesis

It is proposed that an environmental factor acts on a genetically predisposed host.

Infective agents have been accused as the triggers of the autoimmune process e.g. E.coli, enterococci and retroviruses.

There may be a defect in T suppressor cells or a defect in immunoregulation producing damage to the duct system.

Clinical picture

Females are more affected than males especially in their fourth to sixth decades of life.

Pruritus is the usual presenting symptom and is present in 50% of symptomatic patients.

However, the condition may be asymptomatic.

Hepatomegaly and jaundice then develop.

The symptoms may resemble those of obstructive jaundice.

In later stages the patient may have xanthelasmas on the eyelids due to cholesterol deposition.

Malabsorption of fat soluble vitamins and steatorrhea due to deficiency of bile salts may occur in advanced stages.

Osteoporosis is one of the complications.

Diagnosis

Mandatory criteria for diagnosis of PBC:

Clinical and lab evidences of Cholestasis.

M2 antibodies.

Destructive cholangitis on liver biopsy.

Laboratory tests:

Elevated alkaline phosphatase is often the only and the early abnormality in liver biochemistry.

Levels of aminotransferases and GGT are also raised.

Specific autoantibodies are present: Anti Mitochondria' antibodies (AMA) especially M2 are specific and are present in over 95% of patients.

Antinuclear antibodies, smooth muscle antibodies may be present and are not specific. Serum cholesterol is usually raised.

Imaging:

- Abdominal ultrasound is indicated to exclude extrahepatic biliary obstruction and if there is doubt.
- ERCP should be done to make sure that the bile ducts are normal.

Liver biopsy:

Damage and loss of the small bile ducts with ductular proliferation are characteristic of the disease.

N.B: Other causes of hepatic granulomas include; sarcoidosis, tuberculosis, schistosomiasis, and drug reactions.

Treatment

- No effective medical treatment till now.
- Ursodeoxycholic acid (UDCA) improves bilirubin level and liver biochemistry but has no effect on the disease process.
- Supplementation with fat soluble vitamins, and calcium.
- Pruritus is difficult to treat; cholestyramine and naloxone have been tried with mild improvement.
- Control of signs and symptoms of liver cirrhosis and liver cell failure in later stages.
- Liver transplantation is the definite treatment.

N.B: Secondary biliary cirrhosis can result from prolonged (months to years) large duct obstruction by stones or strictures, these are dealt with accordingly,

Primary sclerosing cholangitis (PSC)

It is a chronic cholestatic liver disease characterized by fibrosing inflammatory destruction of both intra and extrahepatic bile ducts.

Pathogenesis

- The exact cause is unknown, Genetic predisposition has been proposed.
- Inflammation of the biliary tract with associated scar formation leads to strictures, ectasias and beading of the wall of the biliary tree.
- These changes are characteristic and are seen on ERCP.
- The changes may range from mild inflammation to established cirrhosis.

Clinical picture

- Males are commonly affected, Age of onset is usually the forties.
- Manifestations of cholestasis are usually present e.g. pruritus and jaundice.
- The course of the disease is usually fluctuating.
- In about 75% of patients inflammatory bowel disease (Crohn's disease and or ulcerative colitis) is present.
- Later on, picture of end stage liver disease eventually takes place.

Investigations

- Elevated bilirubin and alkaline phosphatase are the earliest abnormalities.
- The autoantibody p-ANCA is found in the serum of 60% of patients.
- However, the diagnosis is mainly based on imaging studies; ERCP and/or MRCP show attenuation and the characteristic beading of the extra and intra hepatic biliary radicals.
- Confirmation may need liver biopsy which shows the changes mentioned above.

Treatment

- The only proven treatment is liver transplantation.
- Ursodeoxycholic acid (UDCA) Supplementary treatment as in cases of obstructive jaundice.
- Control of manifestations of end stage liver disease.
- Interventional treatment in the form of endoscopic stents and surgical excision of strictures are of limited value and are mainly used as symptomatic treatment for pruritus.

Hereditary hemochromatosis

Definition

This is an inherited autosomal recessive disease characterized by iron deposition in various organs leading to eventual fibrosis and functional organ failure.

It is due to mutation of HFE gene on the short arm of chromosome 6.

Pathophysiology

The mutant HFE gene interacts with intestinal transferrin receptor leading to loss of normal intestinal control of iron absorption.

Iron is taken up by the intestinal mucosal cells inappropriately exceeding the capacity serum transferrin.

Excess iron is then deposited in the liver and other tissues leading to the damage on the long run.

Early the iron is deposited in periportal hepatocytes, and later it is deposited all over including Kupffer cells, biliary mucosa and connective tissue, Cirrhosis occurs later.

Clinical picture

- Overt clinical manifestations occur in males more than females, this may be explained by the monthly blood loss (and hence iron) in females.
- Affected individuals get the manifestations in their fifties.
- A classic triad: bronze pigmentation of the skin (due to melanin depositing, hepatomegaly and diabetes mellitus secondary to iron deposition in the pancreas.
- Hypogonadism secondary to pituitary affection
- Cardiac manifestations may also be present as heart failure and arrhythmia.
- Thirty percent may develop HCC as a complication of liver cirrhosis.

Investigations

- Elevated serum iron and ferritin with reduction in the TIBC.
- Liver enzymes are usually elevated
- Liver biopsy show increased iron content of the liver and define the extent of the disease.
- MRI detects iron deposition in the pancreas to differentiate the condition from secondary iron overload states (haemosiderosis) in which only the liver is affected.

Treatment

- Venesection to decrease iron load.
- Chelation of iron with desferoxamine.
- Liver transplantation in decompensated cirrhosis, Transplantation will not affect basic disease process (genetic in nature) but will improve quality of life and decrease morbidity.
- Screening of the first degree family members: S iron , ferritin , TIBC & genotyping.

Wilson's disease (hepatolenticular degeneration)

It is a rare autosomal recessive disorder in copper metabolism that results in deposition of copper in various organs mainly; liver, basal ganglia, and cornea.

History of family consanguinity is common.

Pathophysiology

- Dietary copper is absorbed from stomach and upper small intestine, it is transported to the loosely bound to albumin.
- In the liver it is incorporated into ceruloplasmin (a glycoprotein formed in the liver) and secreted in the blood, Copper is normally excreted in bile.
- A defect in the hepatic synthesis of ceruloplasmin leads to failure of hepatic metabolism and excretion.
- Copper is then deposited in the liver leading to progressive damage and in the brain especially basal ganglia leading to extra-pyramidal manifestations, corneal deposition may also occur.

Clinical picture

Presentation may vary from fulminant liver cell failure (commonly in children) to minor laboratory abnormalities in liver function tests.

Finally liver cirrhosis and chronic liver cell failure take place

The following is the classic triad of clinical findings in this disease:

- Hepatomegaly.
- Tremors, dysarthria, and involuntary movements and eventually dementia.
- A specific ocular sign is corneal deposition of copper in the Descemet's membrane known as Kayser-Fleischer ring, it can be seen with close examination or with slit lamp.
- Hemolytic anemia may be present.

Investigations

- Serum copper and ceruloplasmin are reduced.
- Increased 24 urinary copper excretion.
- Liver biopsy will show evidence of acute or chronic inflammation or cirrhosis.
- Measurement of liver copper content is rarely done.
- Slit lamp of the cornea to detect the presence of Kayser-Fleischer ring
- Screening of other family members.
- Hemolytic anemia may be present.

NB: All siblings of patients must be screened: homozygotes must be treated, even if symptom free, heterozygotes do not require treatment.

Alcoholic Cirrhosis

Etiology

- At least 30 gm of alcohol daily for more than 10 years.
- Daily drinking is more hazardous than intermittent drinking.
- Alcohol can enhance the effect of toxic metabolites of drugs (paracetamol) on the liver.

Pathology

- Diffuse micronodular cirrhosis.
- Macrovesicular fatty infiltration.
- Cytoplasmic eosinophilic inclusion bodies (Mallory bodies).

Clinical Picture

- Alcohol can produce a wide spectrum of liver disease: Fatty liver/ alcoholic hepatitis/ alcoholic cirrhosis.
- Clinical manifestations of chronic alcoholism:

GIT: peptic ulcer, gastritis, pancreatitis, malabsorption.

Heart: cardiomyopathy.

Neurological: myopathy, peripheral neuropathy, ataxia, headache, migraine.

Blood: macrocytic anemia, leucopenia, thrombocytopenia.

General: parotid enlargement, Dupuytren's contracture of palmar fascia.

Investigations

- Increased γ GT: it is a sensitive test to determine whether the patient is taking alcohol.
- AST > ALT
- Abdominal U/S, CT, liver biopsy.

Treatment

- Stop alcohol & nutritional support.
- Management of cirrhosis.

Alpha -1 antitrypsin deficiency

- It is a congenital metabolic disease.
- It is due to mutations of the protease inhibitors (Pi) gene.
- α 1 antitrypsin is synthesized in the liver.
- It is an inhibitor of trypsin as well as other proteases e.g. neutrophil elastase.
- Deficiency results in the unopposed action of these enzymes.
- The lung is the major target.
- Damage to the alveoli results in emphysema
- Accumulation of the abnormal α 1 antitrypsin in hepatocytes results in their damage

Clinical Picture

Emphysema is the most common manifestation.

Symptoms begin usually in the third decade of life.

Patient may have had cholestasis & jaundice in childhood (mostly in the first few months of life).

Patients may present with features of chronic hepatitis, cirrhosis or portal hypertension.

Diagnosis

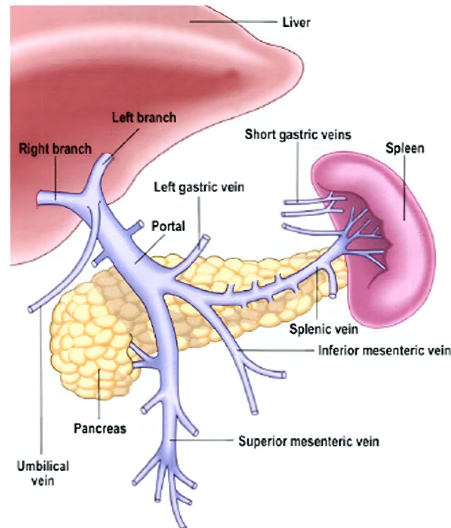
- Low level of α 1 antitrypsin in the serum.
- Gene analysis: M, Z & S phenotypes.

Treatment

- Replacement with synthetic or plasma derived α 1 antitrypsin to treat lung disease.
- Gene therapy.
- Liver transplantation.

Portal hypertension

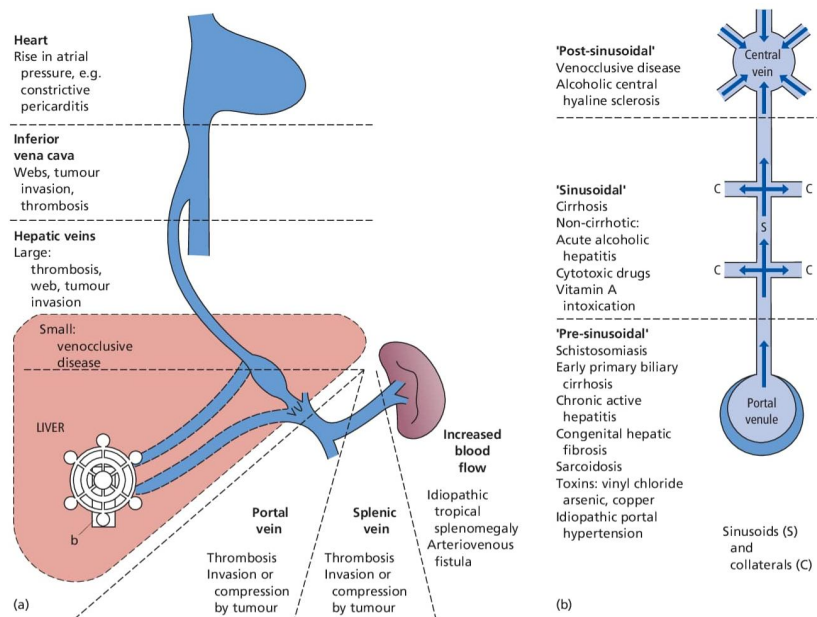
The portal vein is formed by the union of the superior mesenteric and the splenic vein, it enters the liver in two main branches right and left to each lobe at the porta hepatis.



The anatomy of the portal venous system

The portal pressure is normally 5- 10 mmHg , and portal hypertension is considered when the pressure is more than 12 mmHg.

Increased resistance to portal blood flow leads to formation of porto-systemic collaterals which bypass the liver and divert portal blood flow to systemic circulation.



Classification of Portal Hypertension according to the site of obstruction

Causes

Portal hypertension can be classified according to the site of obstruction

1- Prehepatic

Due to blockage of portal vein before the liver

Portal vein thrombosis due to:

- Unknown cause .
- Neonatal sepsis of umbilical vein.
- Abdominal infection .
- Increased blood viscosity: dehydration.
- Malignancy: HCC.
- Congenital narrowing .

2- Intrahepatic

Due to distortion of the liver architecture.

a- Presinusoidal:

- Schistosomiasis due to periportal fibrosis.
- Hodgkin's disease, leukemia, sarcoidosis.
- Congenital hepatic fibrosis
- Toxins: arsenic

b-Sinusoidal and postsinusoidal:

- Cirrhosis (most common)
- Alcoholic hepatitis
- Veno-occlusive disease

3- Post-hepatic

Due to venous blockage outside the liver:

- 1- Budd-Chiari syndrome.
- 2- Right heart failure with tricuspid incompetence.
- 3- Constrictive pericarditis.

The most common cause is: cirrhosis and schistosomiasis in endemic areas like Egypt.

Pathophysiology

When the portal circulation is obstructed, whether it be within or outside the liver and the portal pressure rises above 10-12 mmHg, a remarkable collateral circulation develops to carry portal blood into the systemic venous system (porto-systemic shunts).

The main sites of collaterals are:

1- At the gastro-esophageal junction: Esophageal varices: dilated veins at the lower end of the esophagus which may be associated with gastric varices in the fundus of the stomach. They develop into anastomosis between left and short gastric veins (portal) and azygous vein (systemic).

Esophageal varices are superficial in position and tend to rupture.

2- Rectal varices: due to anastomosis between superior haemorrhoidal (portal) and middle and inferior haemorrhoidal (systemic).

3- At the anterior abdominal wall: due to re-canalization of the umbilical vein (portal) with the superior and inferior epigastric veins (systemic).

4- Collaterals may develop **between the liver and diaphragm**, the spleen and the diaphragm, the spleen and the kidney (lienorenal collaterals) and others.

Clinical picture

- Patients are often asymptomatic and the only clinical evidence is splenomegaly .
- Associated hypersplenism with thrombocytopenia may occur, The spleen is enlarged due to portal hypertension and also due to reticulo-endothelial hyperplasia.
- Hematemesis and melena from rupture of gastro-esophageal varices or congestive gastropathy.
- Encephalopathy due to passage of toxins through porto-systemic shunts.
- Ascites due to associated chronic liver disease.
- Signs of chronic liver disease.

Investigations

- **Upper endoscopy:** to detect the presence of varices (esophageal or gastric), grade of varices, and the presence of active bleeding.

- **Abdominal ultrasound and Doppler study:** It is a noninvasive method to diagnose portal hypertension and possibly to detect its cause.

Diagnosis can be made by measurement of the diameter of the portal and splenic veins, the presence of collaterals (lienorenal, visualize liver, spleen, ascites, etc).

- **Liver function tests**, and liver biopsy to detect the cause of portal hypertension.

Do we need to do upper endoscopy to diagnose esophageal and Gastric varices ?

As stated before , the condition of assessment of hepatic fibrosis and portal hypertension has changed with the development of 2 new definitions

- Compensated advanced chronic liver disease (cACLD)
- Clinically significant portal Hypertension (CSPH).

Patients with cACLD who develop CSPH are at high risk for hepatic decompensation and mortality if left untreated.

The use of elastography in clinical practice has enabled the early identification of patients who are at risk of having CSPH and decompensation.

Patients with a liver stiffness measurement value by transient elastography of > 25 kPa is sufficient to diagnose CSPH are treated with carvedilol to decrease the risk of cirrhosis decompensation.

Variceal hemorrhage

About 90 % of patients with cirrhosis will develop gastro-esophageal varices over 10 years, but only 1/3 of these will bleed.

Esophageal variceal hemorrhage accounts for about 1/3 of deaths in patients with cirrhosis & portal hypertension.

Mortality rate depends on the clinical status of the patient and the severity of the bleeding episode.

Management

Management is divided into:

- Prevention of the first bleeding.
- The active bleeding episode.
- Prevention of rebleeding.

Prevention of the first bleeding

Every patient diagnosed with cirrhosis should have an upper GI endoscopy to assess the possibility of presence of esophageal or gastric varices.

Prophylactic measures to patients with cirrhosis and varices who have not bled include:

Non-selective β -blockers:

- Carvedilol: a non selective beta and alpha1 receptor blocker is used in dose of 6.25 mg twice daily and is well tolerated .

Oral Propranolol, in a dose sufficient to reduce the resting pulse t by 25% (80-160 mg/day) for life.

By using non selective B blocker, the portal inflow is reduced by decrease in cardiac output (β_1) and by blockade of β_2 vasodilator receptor on the splanchnic arteries.

- It is as effective as banding.
- It prevents bleeding from congestive gastropathy and decreases the frequency of rebleeding.

Band ligation through the endoscope is used when there is contraindications or intolerance to β blockers e.g. bronchial asthma, heart block , hypotension.

Management of acute variceal bleeding

1- Resuscitation

- Assess the general condition: pulse-BP.
- Insert IV line and obtain blood for grouping and cross matching : CBC: Hb, PT/INR , liver function tests, creatinine, electrolytes, blood glucose.
- Ryle's tube insertion for aspiration and lavage.
- Restore blood volume: fresh blood transfusion or fresh frozen plasma or plasma expanders, the aim is to maintain systolic blood pressure at 100mmHg.

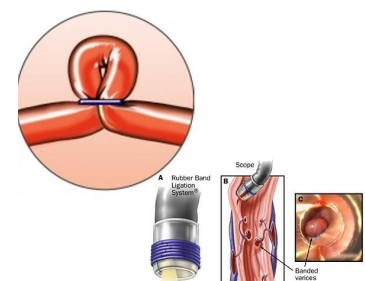
Over transfusion may cause rebleeding. Patient should be kept with a hematocrit value around 30.

2- Urgent endoscopy: only after resuscitation

- To confirm the diagnosis of bleeding varices.
- To exclude other causes of bleeding (e.g. ulcer).
- To detect congestive gastropathy.

Variceal banding or Injection sclerotherapy.

By band ligation ,the varix is sucked into the end of the scope and a rubber band is dislodged over it trip-wire machine that causes necrosis and fibrosis of the varix.



Band ligation of esophageal varices

The varices are injected with a sclerosing agent (like ethanolamine oleate) that produces vessel thrombosis to arrest bleeding.

Gastric varices and big esophageal varices are injected with an adhesive substance (Histoacryl blue) that sticks to the tissues.

Band ligation and Sclerotherapy are the treatment of choice as they arrest bleeding in 80% of cases.

3- Vasoconstrictor therapy: Their main use is for control of emergency bleeding while waiting for endoscopy in combination with endoscopic techniques. They decrease portal inflow by splanchnic arterial constriction.

- Terlipressin: 2 mg/ 6 hours IV then 1mg/4hours for 2-5 days.

The patient complains of abdominal colics, defecation, facial pallor. It should be given to patients with ischemic heart disease.

Somatostatin and somatostatin analogue (octreotide) It has few side effects can be used when there is contraindication to terlipressin.

4- Additional measures:

- Prophylactic antibiotics : oral or IV.
- Measures to prevent encephalopathy.
- Vitamin K IV .
- PPI IV or H2 blockers.
- Sucralfate 1 g/ 6 hours to reduce esophageal ulceration after endoscopic therapy.
- Patients should have nil by mouth until bleeding has stopped.
- Balloon tamponade: This procedure is rarely used now due to its serious complications as aspiration pneumonia but can be used if endoscopic therapy or vasoconstrictive therapy has failed or not available.

Management of an acute rebleed:

- Patients may rebleed within 10 days and in this case endoscopic therapy is repeated.
- **Trans jugular intrahepatic portocaval shunt (TIPS):** TIPS is used when bleeding cannot be stopped or early rebleeding occurs after endoscopic therapy within 5 days. In this technique, a guidewire is passed from the jugular vein into the liver and into the portal vein.

After a balloon expansion of the tract between hepatic and portal vein, an expandable covered metal shunt is placed over the wire to form a channel between the systemic and portal venous systems.

It reduces the hepatic sinusoidal and portal vein pressure by creating a total shunt.

There is an increased risk of portal systemic encephalopathy.

Prevention of recurrent variceal bleeding

Following an episode of variceal bleeding, the risk of recurrence is 60-80% over a 2 years period and this justifies measures to prevent rebleeding

- **Endoscopic therapy** (Band ligation): repeated courses every 2 weeks till obliteration of the varices, Banding is superior to sclerotherapy, then follow up endoscopy.

- Non-selective beta-blockers.

- Surgical porto-systemic shunting shunt: They reduce the risk of rebleeding but produce significant encephalopathy.

The shunts are usually end-to-side portocaval anastomosis or splenorenal shunts (Warren)

- Liver transplantation.

Measurement of the portal pressure

Patency of the portal vein should be assessed before measurement of portal pressure by Doppler ultrasound.

Direct measurement of portal pressure is invasive, expensive and complicated:

- Percutaneous transhepatic measurement.

- Trans-jugular measurement.

- Indirect measurement of portal pressure is safer and less invasive.

- Hepatic vein catheterization: Cannulation of the hepatic vein, balloon occlusion of the hepatic vein and measurement of wedged vein pressure (WHVP). This actually measures the sinusoidal pressure.

- Portal venous pressure gradient (PVPG): difference between portal pressure and pressure in the inferior vena cava.

- Hepatic venous pressure gradient (HVPG): is the difference between the wedged (occluded hepatic venous pressure and the free hepatic venous pressure.

- HVPG has a high correlation with PVPG (when portal hypertension is due to sinusoidal post sinusoidal lesion).

- An HVPG > 12 mmHg is necessary for esophageal varices to form and bleed.

Liver cell failure

Hepato-cellular failure can complicate almost **all forms of liver disease**.

The syndrome **does not complicate portal venous occlusion alone**.

It **may be terminal** in chronic cholestasis, such as primary biliary cirrhosis.

The **overall picture and treatment are similar**, irrespective of the etiology.

Acute liver failure poses special problems.

Principal Functions Of The Liver

1) Formation and secretion of bile.

2) Nutrient and vitamin metabolism:

- ☐ Carbohydrates.
- ☐ Amino acids.
- ☐ Lipids.
- ☐ Water- and fat-soluble vitamins.

3) Inactivation of various substances:

- ☐ Detoxification of ammonia through urea cycle.
- ☐ Toxins.
- ☐ Steroids.
- ☐ Other hormones.

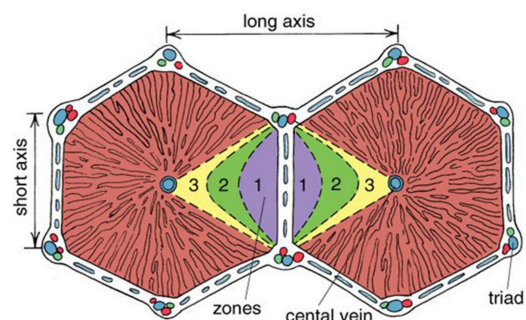
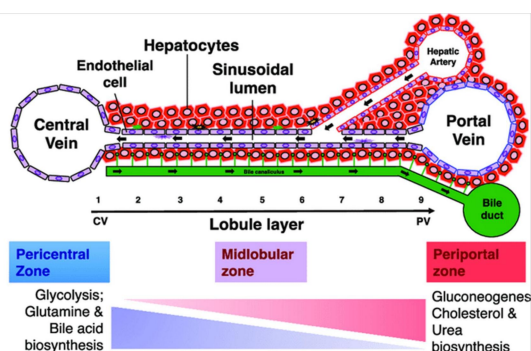
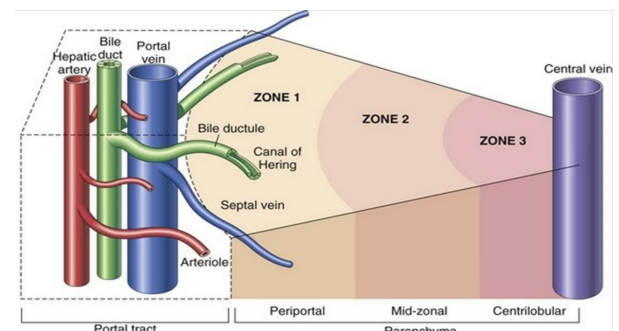
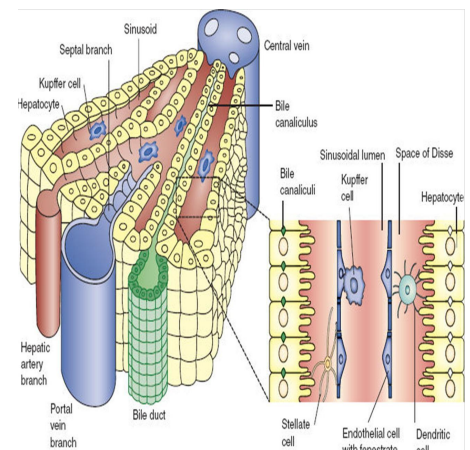
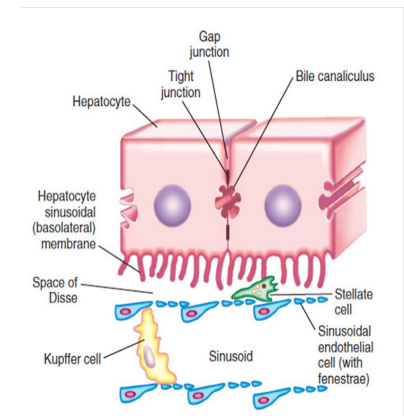
4) Synthesis of plasma proteins:

- ☐ Acute phase proteins.
- ☐ Albumin.
- ☐ Clotting factors.
- ☐ Steroid and hormone binding proteins.

5) Immune function:

- ☐ Kupffer cells.

Microscopic Anatomy Of The Liver



Causes Of Liver Cell Failure

Hepatic Cirrhosis

- 1) **Viral hepatitis** types B $\pm$ delta; C.
- 2) **Alcohol**.
- 3) **Metabolic**, e.g. hemochromatosis, Wilson's disease, α 1-antitrypsin deficiency, non-alcoholic steatohepatitis.
- 4) **Prolonged cholestasis**, intra- and extra-hepatic.
- 5) Hepatic **venous outflow obstruction**.
- 6) Disturbed immunity (**autoimmune hepatitis**).
- 7) **Toxins** and therapeutic agents, e.g. methotrexate, amiodarone.
- 8) **Cryptogenic** cirrhosis

Consequences of cirrhosis

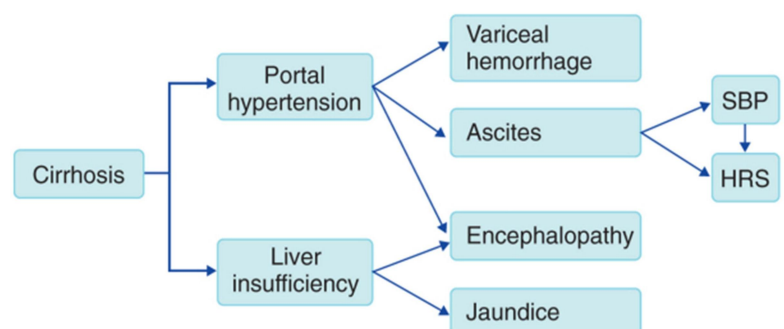
Alteration of hepatic blood flow: Portal hypertension

Reduction in functional cell mass:

- Decreased detoxification: bilirubin, ammonia, drugs.
- Decreased synthesis of: albumin, coagulation proteins, other proteins.

The syndrome of hepatic failure can be viewed as functional rather than anatomical, It comprises some or all of the following features:

- 1) General failure of health.
- 2) Jaundice.
- 3) Hyperdynamic circulation and cyanosis.
- 4) Fever and septicemia.
- 5) Neurological changes (hepatic encephalopathy).
- 6) Ascites.
- 7) Changes in nitrogen metabolism.
- 8) Skin and endocrine changes.
- 9) Disordered blood coagulation.



Ascites

Ascites is free fluid within the peritoneal cavity.

It forms because of conditions **directly involving the peritoneum** (infection, tuberculosis, malignancy), or **diseases remote from the peritoneum** (liver disease, heart failure, hypoproteinemia).

Cirrhosis is the commonest cause, with malignancy, and less frequently cardiac failure and tuberculous peritonitis, being responsible for most other cases.

Causes

Common causes of ascites by the presence or absence of portal hypertension	
Ascites with portal hypertension	Ascites without portal hypertension
<u>Intrahepatic</u> ● Cirrhosis ● Alcoholic hepatitis ● Active chronic hepatitis ● Liver failure ● Veno-occlusive disease ● Liver tumor (primary/metastasis) ● Schistosomiasis ● Sarcoidosis ● Budd-Chiari ● Focal nodular hyperplasia	<u>Peritoneal</u> ● Peritoneal carcinomatosis ● Mesothelioma ● Pseudomyxoma <u>Infectious</u> <u>Mixed</u> ● Myxedema ● Pancreatic ascites, biliary ascites ● Amyloidosis ● Familial Mediterranean fever ● Granulomatous disease
<u>Extrahepatic</u> ● Heart failure, pericarditis ● Portal vein thrombosis	<u>Hypoalbuminemia-related</u> ● Nephrotic syndrome ● Malnutrition ● Bowel disease

Cirrhosis is responsible for 80% of causes of ascites .

Several contributing factors are responsible for ascites formation , they are listed below.

A) Increased hydrostatic pressure:

1. Cirrhosis.
2. Hepatic vein occlusion(Budd-Chiari syndrome).
3. Inferior vena cava obstruction.
4. Constrictive pericarditis.
5. Congestive heart failure.

B) Diminished colloid osmotic pressure:

1. End-stage liver disease with poor protein synthesis.
2. Nephrotic syndrome with protein loss.
3. Malnutrition.
4. Protein-losing enteropathy.

C) Increased permeability of peritoneal capillaries:

1. Tuberculous peritonitis.
2. Bacterial peritonitis.
3. Malignant disease of the peritoneum.

D) Leakage of fluid into the peritoneal cavity:

1. Bile ascites.
2. Pancreatic ascites(secondary to a leaking pseudocyst).
3. Chylous ascites.
4. Urine ascites.

E) Miscellaneous causes:

1. Myxedema.
2. Ovarian disease(Meigs' syndrome).
3. Chronic hemodialysis.

Renin angiotensin physiology

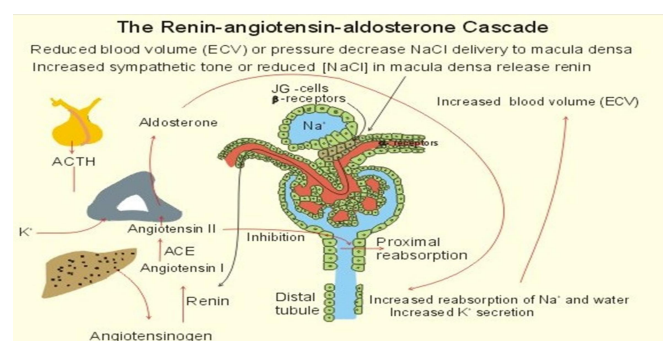
Renin is a **proteolytic enzyme** that is produced and stored in the granules of the **juxtaglomerular cells** surrounding the **afferent arterioles** of glomeruli in the kidney.

Renin acts on the basic substrate **angiotensinogen** made in the **liver** to form the decapeptide **angiotensin I**.

Angiotensin I is then enzymatically transformed by **angiotensin-converting enzyme (ACE)**, which is present in many tissues (particularly the pulmonary vascular endothelium), to the octapeptide **angiotensin II**.

Angiotensin II is a **potent pressor** agent and exerts its action by a direct effect on arteriolar smooth muscle.

In addition, **angiotensin II stimulates production of aldosterone** by the zona glomerulosa of the adrenal cortex



The amount of renin released reflects the combined effects of the following **interdependent** factors:

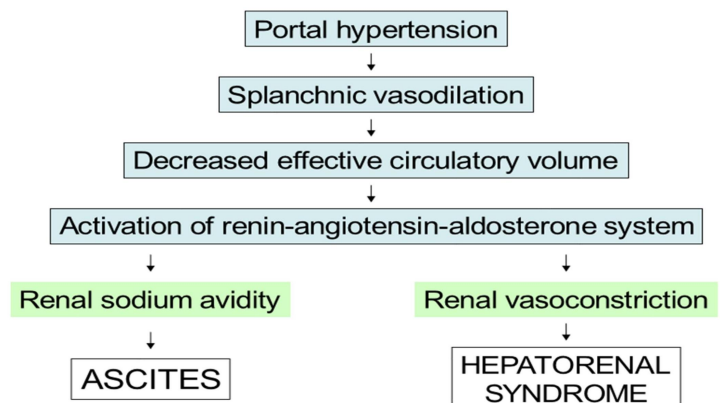
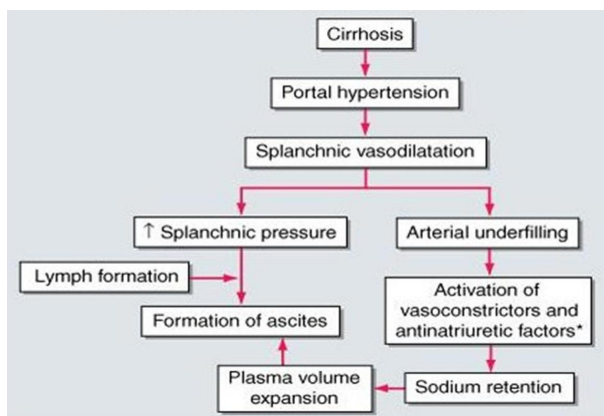
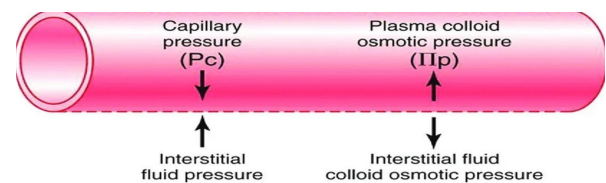
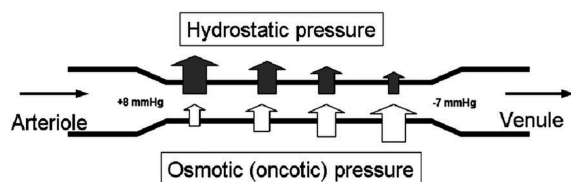
(1) **The juxtaglomerular cells**, which are specialized myoepithelial cells that cuff the afferent arterioles, act as miniature **Pressure Transducers**, sensing renal perfusion pressure.

(2) **The macula densa cells**, a group of distal convoluted tubular epithelial cells directly opposed to the juxtaglomerular cells.

They may function as **Chemoreceptors**, monitoring the sodium (or chloride) load presented to the distal tubule.

(3) **The sympathetic nervous system**, **β receptors** regulates the release of renin in response to assumption of the upright posture.

Pathogenesis of ascites in cirrhosis



Mechanism of ascites formation in liver cirrhosis

- Portal hypertension leads to transudation of fluid in the peritoneal cavity.
- Renal retention of sodium.
- Systemic vascular changes.
- Splanchnic and arterial vasodilatation.
- Increase hepatic lymph production due to congestion of sinusoids.
- Hypoalbuminemia due to decrease hepatic synthesis .

The mechanism involve inappropriate renal sodium and water retention either secondary to vascular changes (underfill and peripheral arterial vasodilatation theory) or as a primary event (overfill theory)

In the underfill theory: stimulation of the rennin-angiotensin-aldosterone system (RAAS) occurs as result of reduced effective circulating volume.

Peripheral vasodilatation → diminished effective arterial blood volume → rise in rennin , aldosterone , vasopressin , and sympathetic stimulation → sodium and water retention and renal vasoconstriction → plasma volume expansion + portal hypertension → Ascites.

In the overfill theory: The liver send a signal which leads to renal sodium and water retention and in the presence of portal hypertension , fluid will accumulates in the peritoneal cavity

Clinical Picture

The onset may be sudden due to a precipitating factor or may be gradual.

Ascites may be graded as follow:

Grade 1: Mild ascites: only detectable by ultrasound examination.

Grade 2: Moderate ascites : manifest by moderate symmetrical distention of abdomen.

Grade 3: Large or gross ascites provokes marked abdominal distension.

Examination:

- Physical signs suggestive of cirrhosis: Jaundice, palmer erythema, abdominal collaterals.
- Abdominal distention & thin limbs "spider man".
- Wide subcostal angle, divarication of recti, umbilicus is shifted downwards, everted ± hernia.
- Scrotal oedema and hydrocele .
- Abdominal wall veins may be apparent due to: caput Medusa (portal hypertension) or due to compression of the IVC
- Palpation: Liver and spleen could be palpated by dipping. Fluid thrill could be felt
- Percussion: Dullness in the flanks, Shifting dullness.
- Secondary effects: Pleural effusion may occurs in 5-10% of patients usually right sided this is due to defect in the diaphragm. hydro-thorax may be seen in the absence of ascites, due to negative intrathoracic pressure during breathing.
- Edema in lower limb: follow ascites, due to hypoalbuminemia.
- Cardiac apex is shifted up & out.
- Neck veins could be seen congested: due to increase intra-pleural pressure & raised diaphragm.

Investigations

- 1) Imaging
- 2) Aspiration

Radiological

Ultrasonography can detect as little as **30 mL of ascitic fluid** with the patient in the right lateral decubitus position, and to detect liver cirrhosis..

Free and loculated collections can be identified.

Computed tomography (CT) scans of the abdomen may detect small amounts of ascites and at the same time evaluate intra-abdominal anatomy, giving important information on the size and state of the liver and spleen and on the presence or absence of varices and tumors.

Characterization of ascitic fluid

A **diagnostic paracentesis** should be performed for new onset ascites or when a complication is suspected.

The procedure is performed under sterile conditions using a 20- to 23-gauge angiocatheter.

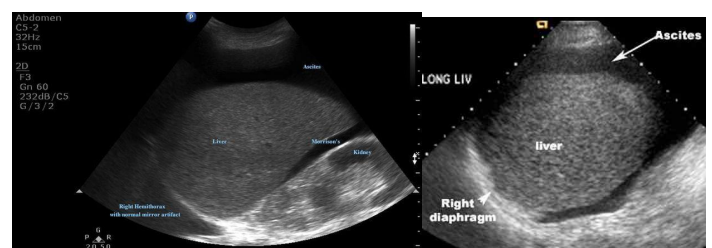
Depending on the clinical situation, fluid may be sent for the following laboratory tests:

1. Gram stain.
2. **Cell count:** endothelial cells/lymphocytes/PNLs/malignant cells. (elevated counts may suggest infection).
3. Bacterial culture.
4. Total protein level: ascitic fluid total protein (AFTP) < 2.5 g/dL.
5. Triglyceride levels (elevated in Chylous ascites).
6. Bilirubin level (may be elevated in bowel perforation).
6. Glucose level.
7. **Albumin level, used in conjunction with serum albumin levels obtained the same day (used to calculate Serum-Ascitic Albumin Gradient [SAAG]) .**

SAAG = Serum albumin — ascitic fluid albumin, In cirrhosis, SAAG is > 1.1g/dL.

8. Amylase level (elevation suggests pancreatic source).
9. Lactate dehydrogenase (LDH) level.
10. Cytology.

Normal US



Serum-Ascitic Albumin Gradient (SAAG)

The gradient correlates directly with portal pressure.

A gradient >1.1 g/dL (high gradient) is characteristic of uncomplicated cirrhotic ascites and differentiates ascites due to portal hypertension from ascites not due to portal hypertension >97% of the time.

Other etiologies of high-gradient ascites include **alcoholic hepatitis, congestive heart failure**, hepatic metastases, **constrictive pericarditis**, and **Budd-Chiari syndrome**.

A gradient <1.1 g/dL (low gradient) suggests that the ascites is not due to portal hypertension with >97% accuracy and mandates a search for other causes such as **peritoneal carcinomatosis, tuberculous peritonitis, pancreatitis**, serositis, pyogenic peritonitis, and nephrotic syndrome.

Differential diagnosis of ascites based on SAAG

Low gradient SAAG < 1.1 g/dl	High gradient ≥ 1.1 g/dl
TB peritonitis	Cirrhosis
Pancreatic Ascites	Fulminant liver failure
Peritoneal carcinomatosis	Alcoholic hepatitis
Biliary ascites	Cardiac ascites
Nephrotic syndrome	Budd-Chiari syndrome

Differential Diagnosis of ascites and its causes:

1- Tuberculous ascites:

- History of TB toxemia.
- Matted intestinal loops.
- Abdominal lymph nodes may be enlarged.
- Ascitic fluid: Ascitic fluid total protein (AFTP) > 2.5 g/dL, SAAG < 1.1 g/dL, excess lymphocytes. Smear & culture for TB. 4- PCR.
- Laparoscopy and biopsy.

2- Malignant ascites:

- Symptoms & signs of primary tumor.
- Ascitic fluid: - AFTP > 2.5 g/dL, SAAG < 1.1 g/dL.
- Malignant cells.
- Increase Lactic acid dehydrogenase.

High Gradient ≥ 1.1 g/dL (11 g/L)

Alcoholic hepatitis
Budd-Chiari syndrome
Cardiac ascites
Cirrhosis
Fatty liver of pregnancy
Fulminant hepatic failure
Massive liver metastases
"Mixed" ascites
Myxedema
Portal vein thrombosis
Sinusoidal obstruction syndrome

Low Gradient < 1.1 g/dL (11 g/L)

Biliary ascites
Bowel obstruction or infarction
Nephrotic syndrome
Pancreatic ascites
Peritoneal carcinomatosis
Postoperative lymphatic leak
Serositis in connective tissue diseases
Tuberculous peritonitis

3- Chylous ascites:

- It may be caused by malignant lymphoma.
- The paracentesis sample is characteristic as it is milky white due to ++ fat & clears with ether.

4- Constrictive pericarditis:

- Increase jugular venous pressure.
- Chest X-ray: calcification.
- ECG & CT of the heart.

5- Budd-Chiari Syndrome:

Increased protein of ascitic fluid.

Abdominal ultrasound: congested veins, caudate lobe is prominent.

Doppler U/S: shows obstruction of hepatic veins.

6- Ovarian tumor:

- Resonance in the flanks, central distension.
- Umbilicus is shifted upwards.

7- Full bladder, pregnancy, flatulence, fatty.

Management of cirrhotic ascites

Bed rest. 70–90mmol sodium diet. Check serum and urinary electrolytes. Weigh daily. Measure urinary volume. Sample ascites
Spironolactone 100–200mg daily
If tense ascites consider paracentesis
After 4 days consider adding frusemide (furosemide) 40mg daily.
Check serum electrolytes
Stop diuretics if pre-coma ('flap'), hypokalaemia, azotaemia or alkalosis
Continue to monitor weight. Increase diuretics as necessary

- Body weight and urinary volume are measured daily.
- Measurement of serum and urinary electrolytes.
- Diagnostic paracentesis.
- Bed rest is advised.
- Salt restriction: This is the main stay of therapy in cirrhotic patients with ascites.

- Diuretics:

1- Aldosterone antagonist, Spironolactone (Aldactone) , amiloride: They block sodium reabsorption in the distal and collecting tubules , and they conserve potassium.

2- Loop diuretic, Furosemide (Lasix): inhibits sodium, potassium and chloride reabsorption from the ascending limb of loop of Henle .

- Two therapeutic approach:

- Spironolactone alone : This can be used alone in mild ascites with a dose of 100 – 200 mg daily.

- Combination therapy Spironolactone 100 mg and Furosemide 40 mg.

Sufficient response means loss of 0.5 Kg/ day or a maximum of 1 Kg /day in the presence of lower limb edema.

If the response is insufficient the dose of Spironolactone and Furosemide can be increased gradually on the expense of development of side effects, maximum of dose of Furosemide is 160 mg /day and of Spironolactone is 400 mg / day.

- **Salt-free albumin I.V** followed by IV Furosemide increases intravascular volume and increases diuresis.

- **Small volume paracentesis:** for < 5 liters removed, I.V. and in this case volume expanders are unnecessary.

- Complications limiting diuretic therapy of ascites

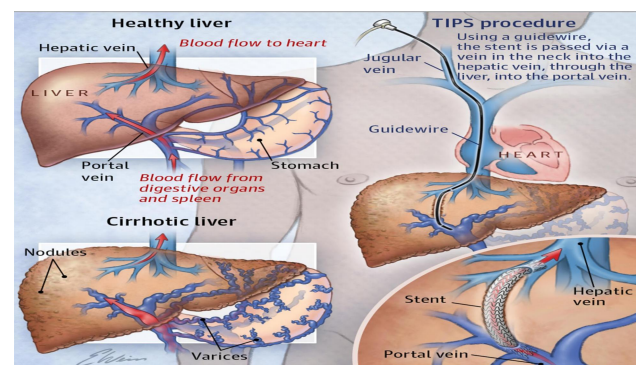
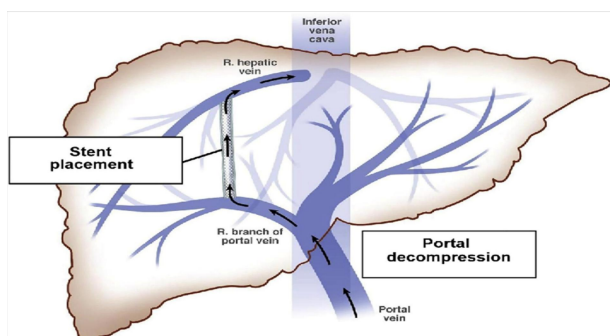
- Development of hepatic encephalopathy, in the absence of any other precipitating factor (It is better to leave the patient wet and wise than dry and demented !).

- Renal impairment: increase of serum creatinine by >100% to a value >2 mg/dl in patients with ascites responding to treatment.

- Hyponatraemia: a decrease of serum sodium by >10 mmol/L to a serum sodium of <125 mmol/L.

- Hypo or hyperkalemia: a change in serum potassium to <3 mmol/L or >6 mmol/L despite appropriate measures.

- Incapacitating muscle cramps.



Transjugular intrahepatic porto-systemic shunt (TIPS)

Treatment options for refractory ascites

Refractory ascites is defined as ascites that cannot be mobilized or prevented from recurring by medical therapy, there is lack of response after dietary salt restriction& intensive diuretic therapy.

It is divided into **diuretic - resistant** (ascites is not mobilized despite maximal diuretic dosage) and **diuretic – intractable ascites** (development of diuretic - induced complications that preclude the use of an effective diuretic dosage)

In 10% to 20% of patients with ascites, the standard medical therapy is ineffective or causes serious side effects

The conditions that may contribute to refractory ascites include **worsening of the primary liver disease**, such as active inflammation, portal or hepatic vein thrombosis, gastrointestinal bleeding, infection, SBP, malnutrition, hepatocellular carcinoma, **superimposed cardiac or renal disease, and hepatotoxic** (e.g., alcohol and acetaminophen) and nephrotoxic substances.

- Large volume (Therapeutic) paracentesis:

The procedure involves the daily removal of **4 to 6 L** of ascitic fluid until the abdomen is completely evacuated, It is safe& effective.

After paracentesis, **adjunctive therapy with diuretics and dietary sodium restriction should be continued** to prevent rapid re-accumulation of ascitic fluid.

Albumin replacement (8gm/ L removed) is more effective in preventing the hypovolaemia and post paracentesis circulatory dysfunction.

The procedure may be repeated at intervals of **2 to 4 weeks**.

Therapeutic paracentesis should be reserved for treatment of tense ascites and ascites that is **refractory** to diuretic therapy.

- Peritoneovenous shunts

- TIPS:

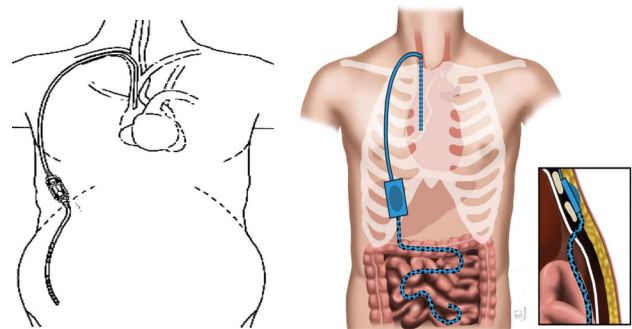
Early experience with TIPS showed a **reduction in diuretic requirements, and a fall in plasma renin and aldosterone activities**.

However, TIPS may precipitate **hepatic encephalopathy** and/or liver failure.

A transjugular intrahepatic portosystemic shunt (TIPS) reduces elevated portal pressure by creating a communication between the **hepatic vein and an intrahepatic branch of the portal vein**.

It used to treat complications of portal hypertension, mainly **variceal bleeding and refractory ascites**, as well as Budd-Chiari syndrome, hepatic hydro-thorax, and **hepatorenal syndrome**.

- Liver transplantation.



Bacterial infection in patients with liver cirrhosis

- Patients with cirrhosis are at risk of developing bacterial infections due to various factors including liver dysfunction, porto-systemic dysfunction, bacterial translocation and immune dysfunction.
- Bacterial infection in cirrhotics should be treated quickly and effectively with potent antibiotic.
- The most common sites for bacterial infection in patients with liver cirrhosis that requires potent empirical antibiotic treatment are:
 - The ascitic fluid (spontaneous bacterial peritonitis)
 - Lung (pneumonia)
 - Urinary tract (pyelonephritis)
 - Skin (cellulitis)

Spontaneous Bacterial Peritonitis (SBP)

It represents a serious complication of ascites with cirrhosis , and it affects 10% of cirrhotic patients.

Infection of the ascitic fluid may be **spontaneous or follow a previous paracentesis**.

The spontaneous type develops in about **8%** of cirrhotic patients with ascites.

It is particularly frequent if the cirrhosis is severely **decompensated**.

Root of infection:

a) Hematogenous spread

b) Translocation of bacteria through the intestinal wall

The infection is blood-borne and in **90% monomicrobial**.

In cirrhotic patients, **bacterial overgrowth** and small intestinal **dysmotility** may contribute.

There is an increased rate of **bacterial translocation** of bacteria across the intestinal wall to mesenteric lymph nodes

Host defenses are abnormal due to decrease ascitic opsonins and abnormal neutrophil function, Reticulo-endothelial function is impaired.

Infection with **more than one organism is likely to be associated with abdominal paracentesis**.

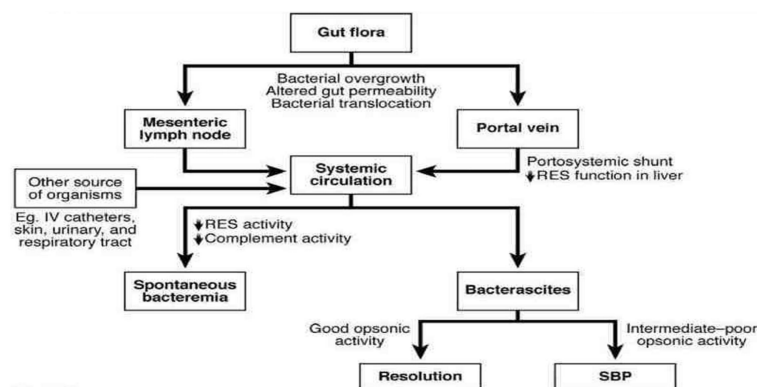
The organisms most commonly found in the infected ascitic fluid belong to the gut flora such as **Escherichia coli**, pneumococci, **Klebsiella** and gram-positive organisms.

Anaerobic infection is rare.

The prognosis has improved over time, but mortality rates remain relatively high (about 20% for SBP).

SBP is an indication to list the patient for liver transplantation.

Pathogenesis of SBP



Clinical Picture & Diagnosis

Spontaneous bacterial peritonitis should be suspected if a patient with known cirrhosis **deteriorates**, particularly with encephalopathy.

Patients with **variceal bleeding or with previous spontaneous bacterial peritonitis** are at particular risk.

Pyrexia, local abdominal pain and tenderness, systemic leucocytosis, nausea, vomiting & may be diarrhea may be noted.

These features, however, may be absent and the diagnosis is made on the index of suspicion with examination of the ascitic fluid.

The patient becomes **drowsy and confused**.

Diagnostic paracentesis: polymorphs in ascitic fluid > 250 cells/mm³.

Ascitic fluid culture.

Prophylaxis

Primary prophylaxis: Patients with cirrhosis and low ascitic fluid protein concentration (<10 g/L) and/or high serum bilirubin levels are at high risk of developing a first episode of SBP, they can be given primary prophylaxis by Norfloxacin 400 mg/day .

Secondary prophylaxis: Norfloxacin 400mg/day for life, after the first attack

The risk of spontaneous bacterial peritonitis is particularly high in cirrhotic patients with **upper gastrointestinal haemorrhage or doing sclerotherapy**.

Oral administration of **Norfloxacin** (400mg/12h for a minimum of 7 days) is currently recommended for this group as well as cases with recurrent peritonitis.

Norfloxacin prophylaxis should be stopped in patients with long-lasting improvement of their clinical condition and disappearance of ascites.

Treatment

Antibiotics should be started empirically.

Albumin has been recommended in association with antibiotics for the treatment of SBP

Patients with spontaneous bacterial peritonitis are particularly at risk of **renal complications** which is probably related to systemic vascular changes.

Third-generation cephalosporins are recommended as first-line antibiotic treatment for community-acquired SBP in countries with low rates of antibiotic resistance.

Five days of parenteral, third-generation cephalosporin such as cefotaxime is usually effective.

A minimal duration of 5 days of treatment is recommended.

Amoxycillin-clavulanic acid is as effective as third generation cephalosporins.

In countries with high rates of antibiotic resistance **piperacillin/tazobactam** or carbapenem should be considered

Antibiotic efficacy should be checked with a second diagnostic paracentesis at 48 hours from starting treatment.

Suspicion of failure of first-line antibiotic if **worsening clinical signs and symptoms** and/or increase, or **no marked reduction in leucocyte count** (at least 25%) in **48 hours**.

Example of antibiotics used in SBP: Cefotaxime 2g/8h IV , ceftriaxone 1g/12 h ,Ciprofloxacin 500mg/12hours

Because of reduced survival, spontaneous bacterial peritonitis is an indication to **consider hepatic transplantation, particularly if recurrent**.

Prognosis: 25% of patients die. 70% recurrence in the first year.

Renal impairment in liver cirrhosis

Hepatorenal syndrome (HRS): It is a progressive, functional liver failure occurring in patients with severe liver disease, liver cirrhosis, portal hypertension and ascites. The kidneys are anatomically and histologically normal and capable of normal function.

Most of these patients have decompensated cirrhosis and tense ascites.

The primary abnormality in the kidneys is **altered blood flow**.

The renal hemodynamic alterations in hepatorenal syndrome are thought to be a consequence of **decreased "effective" blood volume** and **increased sympathetic tone**.

The hallmark of HRS is renal vasoconstriction, although the pathogenesis is not fully understood, however the arterial vasodilatation theory is the most accepted.

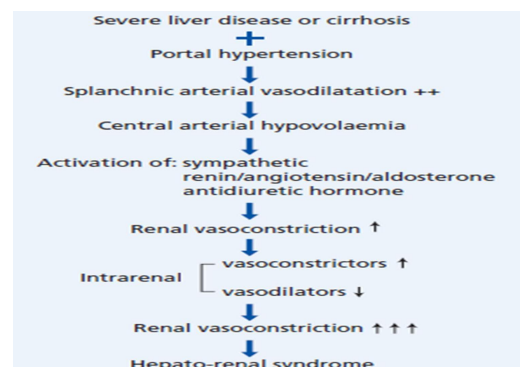
Peripheral and splanchnic vasodilation → Decreased effective blood volume → Activation of RAAS, ADH, sympathetic → Increase Renal vasoconstrictors (rennin, ADH, angiotensin II, sympathetic stimulation) this is associated also with decrease renal vasodilators (PGE2, NO).

Pathogenesis of hepatorenal syndrome

Arterial vasodilatation in the splanchnic and systemic circulation

Renal vasoconstriction

Cardiac dysfunction.



Acute kidney injury (AKI): It is a common complication of advanced cirrhosis, and is predictive of a worse outcome.

- Early recognition of AKI and early identification of the phenotype of AKI are essential to initiate appropriate therapy at an early stage and to improve outcome.
- Patients with advanced cirrhosis have reduced muscle mass (especially women) and decreased creatine production. Thus, patients with serum creatinine within the normal range may have markedly impaired renal function.
- The definitions of AKI in cirrhosis have recently been revised to take into account these difficulties.
- AKI in cirrhosis is now defined by an increase in serum creatinine ≥ 0.3 mg/dL within 48 hours or a percentage increase of serum creatinine $\geq 50\%$ from baseline which is known or presumed to have occurred within the prior 7 days.

Differentiating types of AKI

- AKI is then categorized into 3 stages of increasing severity:
- All types of AKI can occur in patients with cirrhosis (Pre-renal, HRS, acute tubular necrosis (ATN) and post-renal).
- Key point is to differentiate HRS-AKI from ATN.
- Classification of HRS was recently revised.
- HRS-AKI (formerly type I HRS).
- HRS includes renal impairment that fulfills the criteria of HRS but not of AKI (non-AKI-HRS or NAKI) (formerly type II HRS).

Diagnostic criteria for HRS-AKI

- Cirrhosis and ascites.
- No response after 2 consecutive days of diuretic withdrawal and plasma volume expansion with albumin 1 g per kg of body weight.
- Absence of shock.
- No current or recent use of nephrotoxic drugs (NSAIDs, aminoglycosides, iodinated contrast media, etc.).
- No macroscopic signs of structural kidney injury, defined as: Absence of proteinuria (>500 mg/day) and absence of micro haematuria (>50 RBCs per high power field) and normal findings on renal ultrasonography.

Cirrhosis with ascites
Serum creatinine level ≥ 1.5 mg/dL (133 $\mu\text{mol/L}$)
No or insufficient improvement in serum creatinine level (remains ≥ 1.5 mg/dL) 48 hr after diuretic withdrawal and adequate volume expansion with IV albumin
Absence of shock
No evidence of recent use of nephrotoxic agents
Absence of intrinsic renal disease

Deferential diagnosis

Hepatorenal syndrome must be differentiated from **other types of renal failure** that can occur in patients with liver disease.

These include renal failure from exposure to **toxins** such as acetaminophen, NSAIDs, and carbon tetrachloride; infections; **acute tubular necrosis due to hypotension** or exposure to aminoglycoside or angiographic dyes; and **obstructive uropathy**.

Treatment

All patients meeting the current definition of HRS-AKI should be treated with vasoconstrictors and albumin

Besides criteria for the clinical definition, urinary neutrophil gelatinase (NGAL) associated lipocalin, a biomarker of tubular damage, is significantly higher in patients with ATN compared to those with HRS.

A distinction must be made between HRS and acute tubular necrosis (ATN) because vasoconstrictors such as terlipressin are only justified in the presence of HRS.

Terlipressin can be administered by IV boluses (1 mg every 4–6 hours) or by continuous IV infusion (2 mg/day).

- In case of non-response (decrease in SCr <25% from the peak value) after 2 days, the dose of terlipressin should be increased in a stepwise manner to a maximum of 12 mg/day.
- Albumin solution (20%) should be used at 20–40 g/day.
- Serial measures assessing central blood volume can help to titrate the dose of albumin to prevent circulatory overload.
- Noradrenaline can be an alternative to terlipressin.
- Requires a central venous line often in an ICU.
- Midodrine + octreotide can be an option when terlipressin or noradrenaline are unavailable (but efficacy is much lower)

Measures to prevent variceal bleeding (e.g., beta blockers, band ligation)

Pentoxifylline for severe alcoholic hepatitis

Prevention of HRS

Avoid intravascular volume depletion (diuretics, lactulose, GI bleeding, large-volume paracentesis without adequate volume repletion)

Judicious management of nephrotoxins (ACEIs, ARBs, NSAIDs, antibiotics)

Prompt diagnosis and treatment of infections (SBP, sepsis) SBP prophylaxis

Treatment of HRS

Stop all nephrotoxic agents (ACEIs, ARBs, NSAIDs, diuretics) Antibiotics for infections

IV albumin—bolus of 1 g/kg/day on presentation (maximum dose, 100 g daily). Continue at dose of 20–60 g daily as needed to maintain central venous pressure between 10 and 15 cm H₂O

Vasopressor therapy (in addition to albumin):

Terlipressin*—start at 1 mg IV every 4 hr and increase up to 2 mg IV every 4 hr if baseline serum creatinine level does not improve by 25% at day 3 of therapy

OR

Midodrine and octreotide—begin midodrine at 2.5–5 mg orally 3 times daily and increase to a maximum dose of 15 mg 3 times daily. Titrate to an MAP increase of at least 15 mm Hg; begin octreotide at 100 µg subcutaneously 3 times daily and increase to a maximum dose of 200 µg subcutaneously 3 times daily, or begin octreotide at a 25-µg IV bolus and continue at a rate of 25 µg/hr

OR

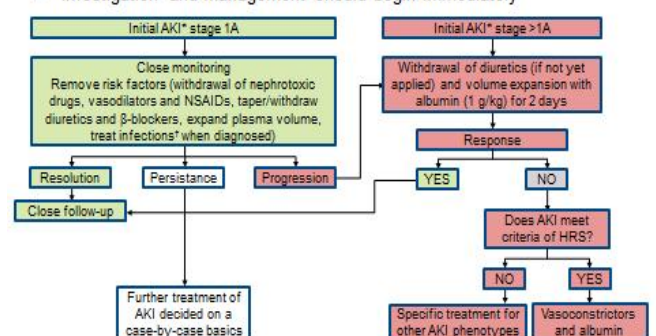
Norepinephrine—0.1–0.7 µg/kg/min as an IV infusion. Increase by 0.05 µg/kg/min every 4 hr and titrate to an MAP increase of at least 10 mm Hg

Duration of vasopressor treatment is generally a maximum of 2 weeks until reversal of hepatorenal syndrome or liver transplantation

Evaluation of patient for liver transplantation

ICA management algorithm for AKI in cirrhosis

- Investigation and management should begin immediately



*Initial AKI stage is defined as AKI stage at the time of first fulfillment of the AKI criteria.
*Treatment of spontaneous bacterial peritonitis should include albumin infusion according to current guidelines.
Adapted from Angelini et al. J Hepatol 2015;62:369–74.
EASL: CPO decompensated cirrhosis. J Hepatol 2016;doi:10.1016/j.jhep.2016.03.024



Jaundice

- Jaundice, from the French jaune (yellow), is the yellow-orange discoloration of the **skin, conjunctivae, and mucous membranes** that results from **elevated concentrations of bilirubin in plasma**.
- Jaundice becomes evident at plasma bilirubin concentrations of **3 to 4 mg/dL**, depending on the patient's normal pigmentation.

Bilirubin Metabolism

Bilirubin production

Bilirubin, a tetrapyrrole pigment, is a breakdown product of **Heme** (ferroprotoporphyrin IX).

About **70–80%** of the 250–300 mg of bilirubin produced each day is derived from the breakdown of **hemoglobin in senescent red blood cells**.

The remainder comes from **prematurely destroyed** erythroid cells in bone marrow and from the turnover of hemoproteins such as myoglobin.

The formation of bilirubin occurs in reticuloendothelial cells, primarily in the **Spleen and Liver**.

The first reaction, catalyzed by the microsomal enzyme **Heme Oxygenase**, oxidatively cleaves the bridge of the porphyrin group and opens the heme ring. The end products of this reaction are **Biliverdin**.

The second reaction, catalyzed by the cytosolic enzyme **Biliverdin Reductase**, reduces the central methylene bridge of biliverdin and converts it to **Bilirubin**.

To be transported in blood, bilirubin must be solubilized. This is accomplished by its reversible, **noncovalent binding to albumin**.

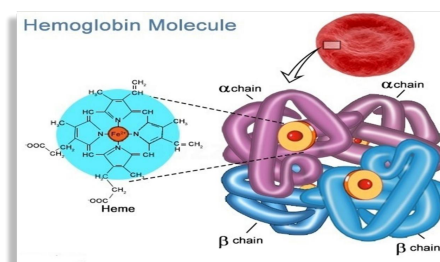
processing of the serum bilirubin load by the hepatocyte occurs in four steps. These are

- 1) **Uptake.**
- 2) **Cytosolic binding.**
- 3) **Conjugation.**
- 4) **Secretion.**

Unconjugated bilirubin bound to albumin is transported to the liver, where it, but not the albumin, is **taken up by hepatocytes**.

After entering the hepatocyte, unconjugated bilirubin is **bound to the cytosolic protein ligandin**.

In the endoplasmic reticulum, bilirubin is solubilized by **conjugation to glucuronic acid**.



The now hydrophilic bilirubin conjugates diffuse from the **endoplasmic reticulum to the canalicular membrane**, where bilirubin monoglucuronide and diglucuronide are actively transported into canalicular bile.

The conjugated bilirubin excreted into bile drains into **the duodenum and passes unchanged through the proximal small bowel**.

Conjugated bilirubin is **not taken up by the intestinal mucosa**.

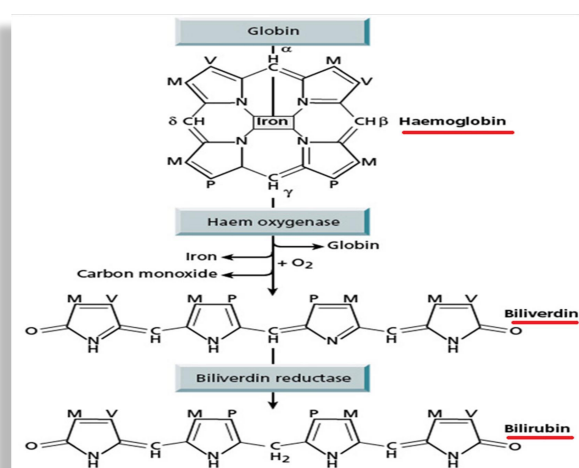
When the conjugated bilirubin reaches the **distal ileum and colon, it is hydrolyzed to unconjugated bilirubin**.

The unconjugated bilirubin is **reduced by normal gut bacteria** to form a group of colorless tetrapyrroles called **Urobilinogens**.

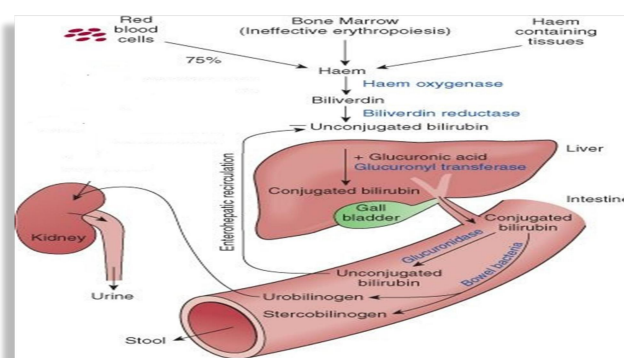
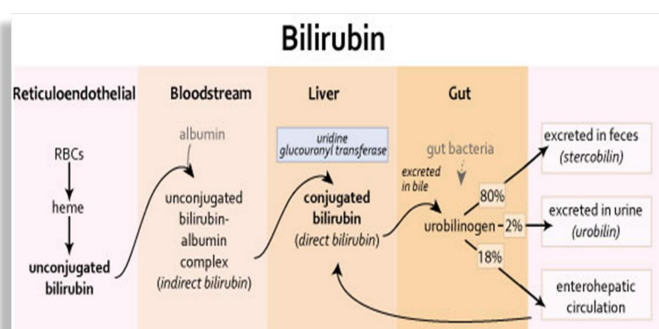
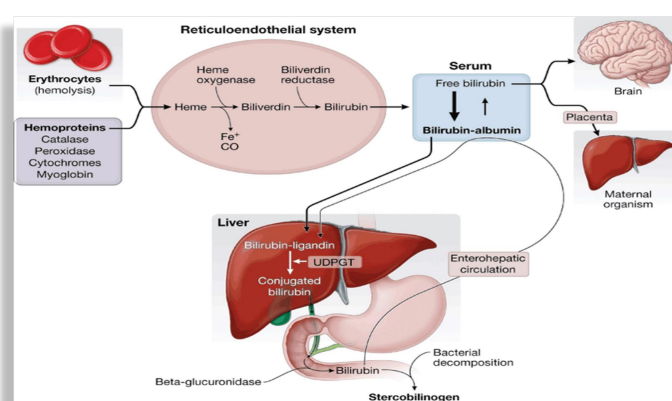
About **80–90% of these products are excreted in feces**, either unchanged or oxidized to **stercobilinogen then stercobilin** which gives stools its dark color.

The remaining 10–20% of the urobilinogens are passively absorbed, enter the portal venous blood, and are re-excreted by the liver.

A small fraction (usually <3 mg/dL) escapes hepatic uptake, filters across the renal glomerulus, and is **excreted in urine**.



Formation Of Bilirubin



Approach to the patient with jaundice

- The bilirubin present in serum represents a balance between **input** from production of bilirubin and **hepatic/biliary removal** of the pigment.

• Hyperbilirubinemia may result from

- (1) **Overproduction** of bilirubin.
- (2) **Impaired** uptake, conjugation, or excretion of bilirubin.
- (3) **Regurgitation** of unconjugated or conjugated bilirubin from **damaged hepatocytes or bile ducts**.

• In other words hyperbilirubinemia and jaundice may result from:

- A) **Isolated disorders** of bilirubin metabolism.
- B) **Liver** disease.
- C) **Obstruction** of the biliary tract.

• An increase in unconjugated bilirubin in serum results from either:

- 1) Overproduction
- 2) impairment of uptake, or conjugation of bilirubin.

• An increase in conjugated bilirubin is due to:

- 1) decreased excretion into the bile ductules.
- 2) backward leakage of the pigment.

The initial steps in evaluating the patient with jaundice are to determine:

- (1) whether the hyperbilirubinemia is predominantly conjugated or unconjugated in nature (fractionation of Bilirubin).
- (2) whether other biochemical liver tests are abnormal.

Differential Diagnosis of Jaundice and Hyperbilirubinemia

Isolated Disorders Of Bilirubin Metabolism

Unconjugated Hyperbilirubinemia

Increased bilirubin production

Examples: hemolysis.

Decreased hepatocellular uptake

Examples: drugs (e.g., rifampin).

Decreased conjugation

Examples: Gilbert's and Crigler-Najjar syndromes.

Conjugated Or Mixed Hyperbilirubinemia

Decreased canalicular transport

Examples: Dubin-Johnson syndrome.

Gilbert's Syndrome

- It is defined as benign, **familial, mild, unconjugated hyperbilirubinemia** not due to hemolysis and with normal routine tests of liver function and hepatic histology.
- **Jaundice is mild** and intermittent.
- Deepening may follow an **intercurrent infection or fasting** and is associated with malaise, nausea and often discomfort over the liver.
- Patients with Gilbert's syndrome have a **deficiency in hepatic bilirubin glucuronidation**.
- **Hyperbilirubinemia is life long** and not associated with increased morbidity.

Hepatocellular Conditions That May Produce Jaundice

Viral Hepatitis

Hepatitis A, B, C, D, and E

Epstein-Barr virus

Cytomegalovirus

Alcohol

Drug toxicity

Environmental Toxins: Vinyl chloride

Wilson's disease

Autoimmune hepatitis

Cholestatic Conditions That May Produce Jaundice

- Cholestasis is defined as the **failure of normal bile to reach the duodenum**.
- This may be due to pathology anywhere between the **hepatocyte and the ampulla of Vater**.
- The term '**obstructive jaundice**' is **not used**, as in many instances no mechanical block can be shown in the biliary tract.

Syndrome Of Cholestasis

- Cholestasis is **interference with bile flow or formation**.
- This can occur anywhere between the hepatocyte and the ampulla of Vater.
- **Functionally**, cholestasis is defined as a decrease in canalicular bile flow.
- **Morphologically**, cholestasis is defined as the accumulation of bile in liver cells and biliary passages.
- **Clinically**, cholestasis is the retention in the blood of all substances normally excreted in the bile.
- The importance of the distinction between extra and intra-hepatic cholestasis is that **symptoms and biochemistry may not separate them**.
- There is a need for a diagnostic algorithm to differentiate between the two.

Classification

Extra-hepatic Cholestasis	Intra-hepatic cholestasis
• Encompasses conditions where there is physical obstruction to the bile ducts. • Usually this is outside the liver.	• Includes those conditions where there is no demonstrable obstruction.

Intrahepatic Conditions Producing Cholestasis

Viral hepatitis

Fibrosing cholestatic hepatitis—hepatitis B and C

Hepatitis A, Epstein-Barr virus

Alcoholic hepatitis

Drug toxicity

Pure cholestasis—anabolic and contraceptive steroids

Cholestatic hepatitis—chlorpromazine

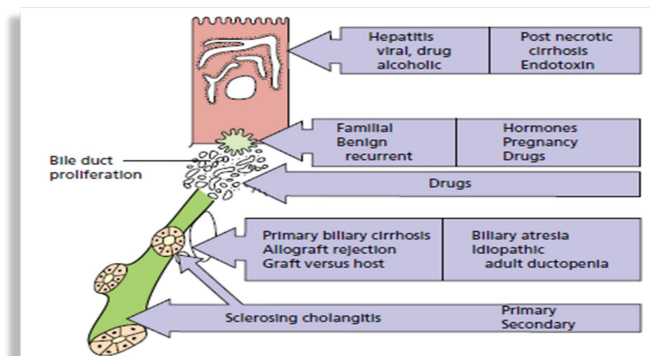
Chronic cholestasis—chlorpromazine

Primary biliary cirrhosis

Primary sclerosing cholangitis

Cholestasis of pregnancy

Benign postoperative cholestasis



Extrahepatic Conditions Producing Cholestasis

Malignant

Cholangiocarcinoma

Pancreatic cancer

Ampullary cancer

Malignant involvement of the porta hepatis lymph nodes

Benign

Choledocholithiasis

Postoperative biliary structures

Chronic pancreatitis

Clinical features of cholestasis

- Prominent features of cholestasis, both acute and chronic, are **itching and malabsorption**.
- **Bone disease** (hepatic osteodystrophy) and **cholesterol deposition** (xanthomas, xanthelasmas) are seen with chronic cholestasis, which is also associated with **skin pigmentation**.
- In contrast to the patient with hepato-cellular disease where there is malaise and physical deterioration, **the cholestatic patient feels well**.
- On examination, **the liver is usually enlarged**.
- **Splenomegaly is unusual** except in biliary cirrhosis where portal hypertension has developed.
- **Steatorrhea** is proportional to the degree of jaundice. It is due to the lack of sufficient intestinal bile salts for the absorption of dietary fat and fat-soluble vitamins.
- **Hepatic osteodystrophy**: Bone disease is a complication of chronic liver disease and in particular chronic cholestasis.
- **Bone pain** and fractures occur.
- Possible mechanisms are **osteomalacia and osteoporosis**.
- Bone disease manifests as **loss of height, back pain** (usually mid-thoracic or lumbar), **collapsed vertebrae** and fractures particularly of ribs with minimal trauma.

Extra-hepatic Effects Of Cholestasis

- Itching and jaundice are self-evident, but there are **numerous other less obvious** effects of cholestasis.
- **Cardio-vascular** responses are abnormal and peripheral vasoconstriction in response to hypotension is impaired.
- The **kidneys** have an increased susceptibility to hypotension and hypoxic damage.
- The **gastric mucosa** is more susceptible to ulceration
- Thus although deeply jaundiced patients with cholestasis may appear well apart from itching, there are **metabolic and functional changes** that under the stress of surgical and non-surgical procedures may result in: **acute renal failure, haemorrhage, wound dehiscence and an increased risk of sepsis**.

Laboratory findings in cholestasis

- All the constituents of the bile show an increased level in the serum. Conjugation of biliary substances is intact but excretion defective
- The **serum conjugated bilirubin** level is raised
- The serum **alkaline phosphatase** level is raised, usually to more than three times the upper limit of normal
- Serum **γ glutamyl- transpeptidase** levels are raised.
- Serum **albumin and globulin** concentrations are normal.
- The **serum aspartate transaminase** is usually less than 100iu/l.
- **Urine:** Conjugated bilirubin is present.

Clues To The Differential Diagnosis Of Jaundice: Biliary Obstruction Versus Liver Disease

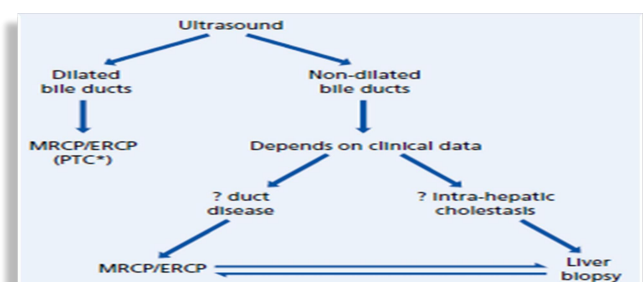
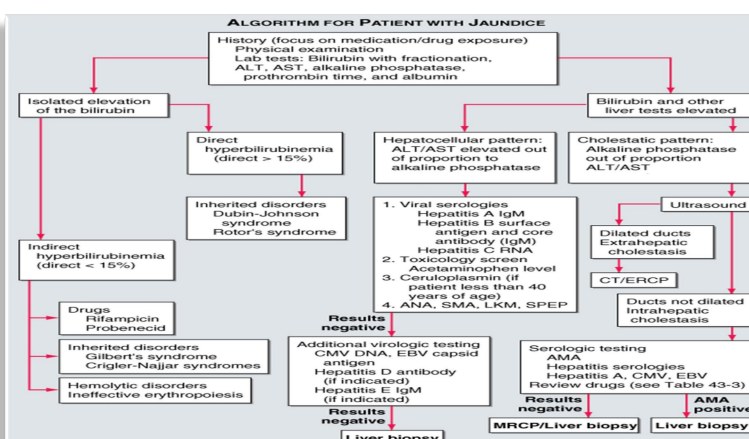
Parameter	Biliary Obstruction	Liver Disease
History	Abdominal pain Fever, rigors Prior biliary surgery Older age	Anorexia, malaise, myalgias (viral prodrome) Known viral exposure History of blood product receipt or injection drug use Exposure to known hepatotoxin Family history of liver disease
Physical examination	Fever Abdominal tenderness Palpable abdominal mass Abdominal surgical scar	Spider telangiectasias Stigmata of portal hypertension (e.g., prominent abdominal veins, splenomegaly, ascites) Asterixis
Laboratory studies	Predominant elevation of serum alkaline phosphatase relative to aminotransferases* Prothrombin time (INR) normal or normalizes with vitamin K administration Leukocytosis Elevated serum amylase or lipase level	Predominant elevation of serum aminotransferase levels relative to alkaline phosphatase Prolonged prothrombin time that does not normalize with vitamin K administration Thrombocytopenia Serologies indicative of specific liver disease

Liver Function Tests

Test significance	Individual test
Detoxification and excretory functions.	Serum bilirubin. Urine bilirubin. Serum ammonia.
Indicators of liver cell injury.	AST, ALT.
Cholestasis.	Alkaline phosphatase, GGT, 5 nucleotidase.
Synthetic functions.	Serum albumin, coagulation factors.

Causes of isolated elevation of alkaline phosphatase

- ☐ Primary or metastatic hepatic tumors.
- ☐ Granulomatous liver diseases.
- ☐ Primary biliary cirrhosis.
- ☐ Non-Hodgkin lymphoma.
- ☐ Hyperthyroidism.
- ☐ Any bone diseases with rapid turnover.
- ☐ Sclerosing cholangitis.



Approach To Patient With Laboratory Proved Cholestasis

Hepatic encephalopathy

Hepatic encephalopathy is a **reversible neuropsychiatric state that complicates liver disease**.

A **spectrum of syndromes** exists, and it may accompany various forms of liver diseases.

Types of encephalopathy

Type of encephalopathy	% Survival	Aetiological factors
Acute liver failure	20*	Viral hepatitis Alcoholic hepatitis Drug reactions and overdose
Cirrhosis with precipitant	70–80	Diuresis Haemorrhage Paracentesis Diarrhoea and vomiting Surgery Alcoholic excess Sedatives Infections Constipation
Chronic portal-systemic encephalopathy	100	Portal-systemic shunting Dietary protein intake Intestinal bacteria



Flapping tremors

Clinical features

The picture is complex and **affects all parts of the brain**.

There are **neurological and psychiatric** components.

For descriptive purposes, features of encephalopathy can be separated into **changes in consciousness, personality, intellect and speech**.

Disturbed consciousness:

Hypersomnia appears early and progresses to reversal of the normal sleep pattern.

Further deterioration results in reaction only to intense or noxious stimuli.

Personality changes:

These include childishness, irritability and loss of concern for family.

Intellectual deterioration

Varies from slight impairment of organic mental function to gross confusion.

Speech is slow and slurred and the voice is monotonous.

The most characteristic neurological abnormality is the **'flapping' tremor (asterixis)**.

This is due to **impaired inflow of joint and other afferent information** to the brainstem reticular formation resulting in **lapses in posture**.

Sometimes **arms, neck, jaw, protruded tongue**, retracted mouth and tightly closed eyelids are involved, and **the gait is ataxic**.

Precipitants Of Acute Hepatic Encephalopathy In The Cirrhotic Patient

Electrolyte imbalance

- a) Diuretics
- b) Vomiting
- c) Diarrhoea

Bleeding

- A. Oesophageal and gastric varices
- B. Gastro-duodenal erosions
- C. Mallory–Weiss tear

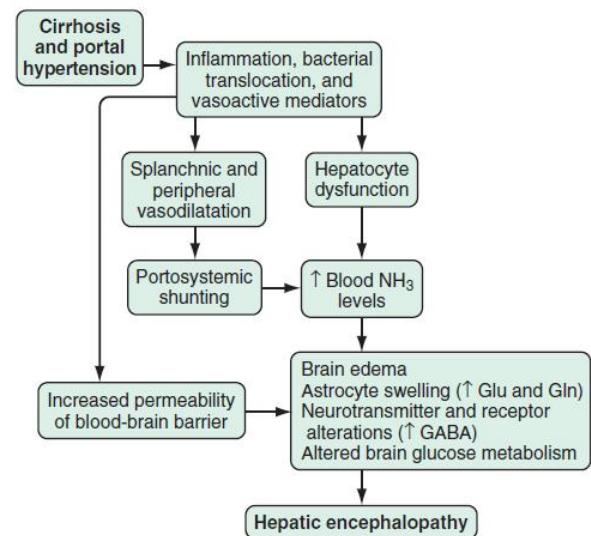
Drugs

Infection

- a. Spontaneous bacterial peritonitis
- b. Urinary
- c. Chest

Constipation

Large protein meal



Proposed pathophysiology of hepatic encephalopathy. GABA, γ -aminobutyric acid; Gln, glutamine; Glu, glutamate; NH_3 , ammonia.

Pathogenetic mechanisms

- 1) **Failure of hepatic clearance** of gut derived substances, either through hepatocellular failure or shunting.
- 2) **Altered amino acid metabolism**, both of which result in changes in **cerebral neurotransmission**.
- 3) **Ammonia** is thought to play an important role; other **neurotransmitter systems** are strongly implicated.

Human body **cannot easily and quickly remove ammonia** usually have to convert it to some other substance, like urea or uric acid, which are much less toxic

The urea cycle (also known as the **ornithine cycle**) is a cycle of biochemical reactions occurring in many animals that **produces urea from ammonia**.

In mammals, the urea cycle takes place **primarily in the liver, and to a lesser extent in the kidney**.

Ammonia and glutamine:

Ammonia is produced from the **breakdown of proteins**, amino acids, purines and pyrimidines.

About half of the ammonia arising from the intestine is synthesized by **bacteria**, the remainder coming from **dietary protein and glutamine**.

The liver normally converts **ammonia to urea and glutamine** through the urea cycle.

In hepatic encephalopathy **blood ammonia levels are elevated** in 90% of patients, Brain levels are also increased.

Plasma **aromatic amino acids**, tyrosine, phenylalanine and tryptophan, are **increased** in patients with liver disease probably due to **failure of hepatic deamination**.

The **branched-chain amino acids**, valine, leucine and isoleucine, are **decreased**.

This might be due to **increased metabolism by skeletal muscle and kidneys**, secondary to the hyperinsulinemia of chronic liver disease.

The imbalance in plasma levels **allows more aromatic amino acids to pass an abnormal blood-brain barrier**.

An **increase in phenylalanine level in the brain leads to inhibition of dopa production** and the formation of false neurotransmitters.

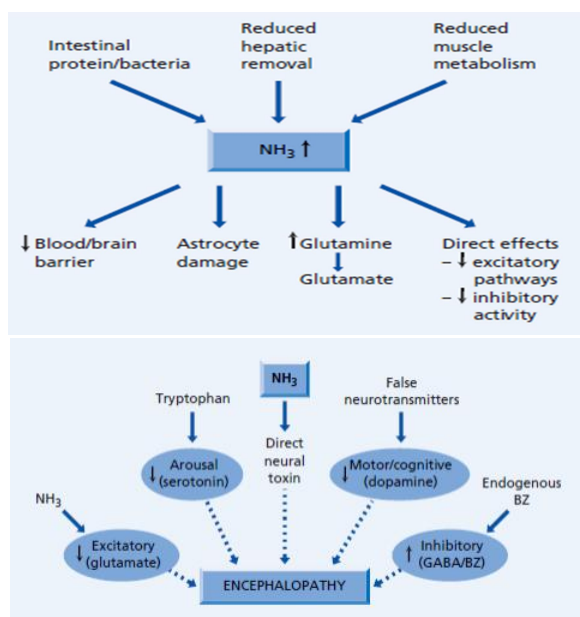
γ Aminobutyric acid (GABA)

GABA is the principal **inhibitory neurotransmitter** in the Brain.

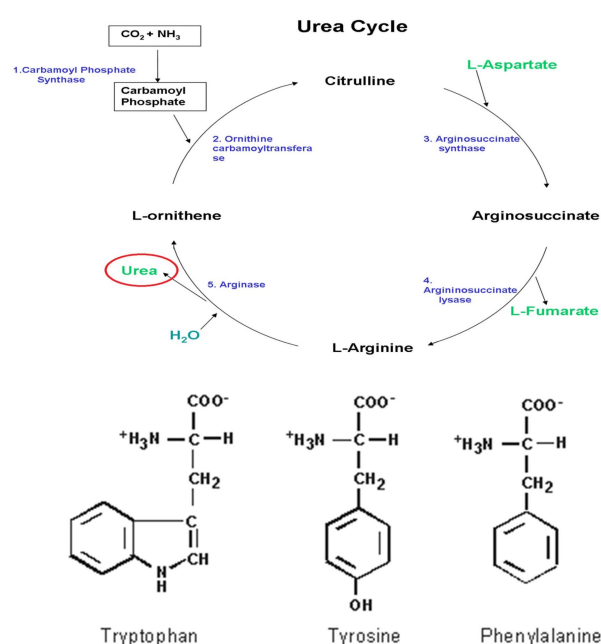
GABA is **synthesized by gut bacteria**, and that entering the portal vein is **metabolized by the liver**.

In the presence of liver failure or portal-systemic shunting it **enters the systemic circulation**.

There are **increased GABA levels** in the plasma of patients with liver disease and hepatic encephalopathy.



By Wilson



Treatment of hepatic encephalopathy

Treatment strategies

Identification and treatment of the precipitating cause

Intervention to reduce the production and absorption of gut derived ammonia and other toxins

Prescription of agents to modify neurotransmitter balance

Diet

In the acute attack dietary **protein is reduced** to 20 g/day.

In patients after an acute episode of coma, a normal protein intake is soon achieved.

It is important in **cirrhotic patients to avoid protein restriction** for any longer than is necessary since these patients have a higher than normal protein requirement (1.2 g/kg/day) to remain in positive balance.

Vegetable protein may be used. The latter is less ammoniagenic and contains small amounts of methionine and aromatic amino acids.

Antibiotics

Neomycin is a non absorbable aminoglycoside given orally in doses of 2 to 6 g per day.

Metronidazole has become the oral antibiotic of choice for acute management of hepatic encephalopathy when lactulose alone does not achieve adequate results.

Rifaximin (Xifaxan) has recently shown to be effective in the treatment of PSE in the dose range of 400 mg 2-3 times daily. It is a non absorbable, gut-specific antibiotic that alters the intestinal flora with any systemic side effects.

Lactulose and lactitol

The human intestinal mucosa does not have an enzyme to **split these synthetic disaccharides**.

When given by mouth lactulose **reaches the caecum where it is broken down by bacteria predominantly to lactic acid**.

The **osmotic volume** of the colon is increased and **the faecal pH drops**.

The colonic **fermentative bacteria prefer lactulose to blood** when both are present.

The growth of **lactose-fermenting organisms is favored** and organisms such as bacteroides, which are **ammonia formers, are suppressed**.

Faecal acidity would **reduce the ionization and hence absorption of ammonia** (also amines and other toxic nitrogenous compounds).

Side-effects include flatulence, diarrhea and intestinal pain.

Purgation

Lactulose or lactose enemas may be used and are superior to water

Sodium benzoate and L-ornithine -L-aspartate

Sodium benzoate **promotes urinary excretion of ammonia** and is as effective as lactulose and is less expensive.

L-ornithine-L-aspartate treatment **promotes hepatic removal of ammonia** by stimulating residual hepatic urea cycle activity and promoting glutamine synthesis, particularly in skeletal muscle.

Levodopa and bromocriptine

If portal-systemic encephalopathy is related to a **defect in dopaminergic neurotransmission** then replenishment of cerebral dopamines should be beneficial

Dopamine does not pass the blood–brain barrier, but its precursor, levodopa, does and can cause temporary arousal in acute hepatic encephalopathy.

Bromocriptine is a specific dopamine receptor agonist with a prolonged action.

Branched-chain amino acids

In cirrhotic patients the **serum branched chain amino acids valine, leucine and isoleucine are low**, and aromatic amino acid levels are increased.

Infusions of solutions containing a high concentration of branched-chain amino acids have been used to treat acute and chronic hepatic encephalopathy.

Hepatic transplantation

This may be the ultimate answer to the problem of chronic hepatic encephalopathy

Acute liver failure (Fulminant Hepatitis)

Acute liver failure describes the clinical syndrome of severe impairment of liver function (**Encephalopathy, Coagulopathy And Jaundice**) **Within 6 Months Of The Onset Of Symptoms**.

Coagulopathy and Encephalopathy are the twin cardinal features that reflect the severity of liver injury, and both are required to make the clinical diagnosis of ALF.

The vast majority of patients have **no evidence of preexisting liver disease**, but 2 exceptions to this requirement have been made:

- (1) an acute presentation of **Wilson disease** in a young adult that presents in the setting of previously asymptomatic cirrhosis.
- (2) **Reactivation of HBV infection**, either spontaneously or as a consequence of immunosuppression or chemotherapy.

Classification

Classification of acute liver failure

Liver failure subcategory	Jaundice to encephalopathy	Clinical presentation	Common aetiologies	Prognosis
Hyperacute	0–7 days	Cerebral oedema common	Paracetamol, hepatitis A, ischaemia	Fair
Acute	8–28 days	Cerebral oedema less common	Hepatitis B, drugs	Poor
Subacute	29 days to 12 weeks	Cerebral oedema rare; ascites, peripheral oedema and renal failure more common	Drugs, indeterminate	Very poor

Causes

Causes of acute liver failure

Infections
Hepatitis A, B, C, D, E
Herpes simplex
Epstein–Barr virus
Cytomegalovirus
Transfusion-transmitted virus (TTV)
Dengue fever
Drugs and toxins
Paracetamol
Carbon tetrachloride
Idiosyncratic drug reactions*
Mushroom poisoning
Sea anemone sting
Ischaemic
Cardiogenic shock
Hypotension
Heat stroke
Cocaine, methamphetamines, ephedrine
Vascular
Acute Budd–Chiari syndrome
Sinusoidal obstruction syndrome
Miscellaneous
Wilson's disease
Acute fatty liver of pregnancy
Eclampsia/ HELLP syndrome
Malignancy
Primary graft non-function after liver transplantation

Clinical Features And Diagnosis

The diagnosis of ALF is clinical and based on the presence of **encephalopathy** in a patient with acute liver injury manifested by **coagulopathy**.

The encephalopathy is usually overt and ranges from **confusion to coma**.

1) Encephalopathy

Encephalopathy is mandatory for a diagnosis of ALF. It is classically graded on a scale of 1 to 4.

2) Intracranial Hypertension and Cerebral Edema

Cerebral edema has been a hallmark complication of ALF, **major cause of death**.

Cerebral blood flow is disturbed and generally increased in patients with acute liver failure; persistent **vasodilatation and loss of auto-regulation** may generate hyperemia, and the consequent augmentation in cerebral blood volume may exacerbate brain edema.

The clinical features of cerebral edema include **systemic hypertension, decerebrate posturing, hyperventilation, abnormal pupillary reflexes, and ultimately impairment of brainstem reflexes and functions**.

3) Hemodynamic Changes and Circulatory Failure

The initial profile is of a **hyperdynamic circulation**, increased cardiac output, and reduced systemic peripheral vascular resistance.

Progressive disease leads to **circulatory failure** either as a result of a falling cardiac output or an inability to maintain an adequate mean arterial pressure.

This is a common cause of death in ALF.

4) Infection

Patients with ALF are notoriously prone to infection, The predominant bacteria were **Staphylococcus aureus and Streptococci**.

5) Renal Failure

In most non-acetaminophen cases, renal failure tends to be a **consequence of liver failure and develops later in the course of the disease**.

6) Hematologic Abnormalities

In addition to **decreased synthesis of coagulation factors** by the liver, there is evidence of increased peripheral consumption.

Quantitative and qualitative defects in **platelet function** are well described in ALF.

Diagnostic Studies

Haematology

Complete blood count: white blood cells, haemoglobin, haematocrit, anti-hepatitis B core
Coagulation panel: prothrombin time/INR, factor V
Blood group

Biochemical

Serum chemistries: sodium, potassium, bicarbonate, chloride, urea, creatinine, calcium, magnesium, phosphate, glucose
Hepatic panel: aspartate aminotransferase (AST), alanine aminotransferase (ALT), alkaline phosphatase, albumin, total protein, total bilirubin

Arterial blood gas

pH, PaCO₂, PaO₂, lactate, ammonia

Virology

Hepatitis B surface antigen and IgM anticore
Hepatitis A (IgM) antibody
Hepatitis C antibody (for underlying chronic infection)
HCV RNA
IgM hepatitis E antibody (in endemic areas)
Hepatitis D antibody if hepatitis B positive
HSV, CMV, EBV PCR (if history of immunosuppression)
Human immunodeficiency virus (if considering transplantation)

Autoimmune markers

Antinuclear antibody (ANA)
Antismooth-muscle antibody (SMA)
Antiliver/ kidney microsome 1 (ALKM1)
Immunoglobulins

Toxicology

Paracetamol level
Blood alcohol
Urine drug screen

Miscellaneous

Urine copper
Pregnancy test

Microbiology

Blood culture, aerobic and anaerobic
Urine culture and microscopy
Sputum culture and microscopy

Other studies

Chest X-ray, electrocardiogram
Liver ultrasound with Dopplers
Electroencephalogram (EEG), non-contrast head CT (in some cases)
Liver biopsy

Management

1) Neurologic Complications

Management of the porto-systemic encephalopathy of cirrhosis has centered on lactulose and non – absorbable antibiotics. However, **lactulose has shown no benefit in acute liver failure.**

There is not enough evidence to recommend non - absorbable antibiotics in acute liver failure.

In cirrhosis, **L-ornithine and L-aspartate (LOLA)** treats encephalopathy by increasing muscle metabolism of ammonia but a recent randomized controlled trial showed no benefit in acute liver failure.

Therefore, the management of the encephalopathy is essentially that of the **underlying liver disease.**

2) Treatment of cerebral edema

Mannitol (0.25 – 1 g/kg i.v. bolus): use for severe elevation of ICP or first clinical signs of herniation, monitor urine output and serum osmolality.

Hypertonic saline (30% 5 – 10 mL/h): goal serum sodium 145 – 155 mmol/L, avoid rapid correction.

Hyperventilation: effects short - lived; may use for impending herniation.

3) Infection

Aseptic techniques.

Surveillance for and **prompt antimicrobial treatment** of infection required.

Anti-fungal coverage for patients not responding to broad spectrum antibiotics.

Antibiotic prophylaxis possibly helpful but not proven.

4) Coagulopathy

Vitamin K (10 mg i.v. or subcutaneous).

FFP: give only for invasive procedures or active bleeding.

Platelets: give for platelet counts < 10 000/mm³ or invasive procedures.

Recombinant activated factor VII (40 µg/kg bolus): give 30 min before procedure if coagulopathy refractory to FFP.

Cryoprecipitate: for fibrinogen < 100 mg/dL and bleeding.

Exchange plasmapheresis: allows transfusion of large amount of FFP.

Prophylaxis for stress ulceration: give H2 blocker or PPI.

5) Hemodynamics/ renal failure

Arterial line.

Pulmonary artery catheterization.

Volume replacement.

Check cortisol and cosyntropin stimulation test; hydrocortisone 200 – 300 mg/day for **adrenal insufficiency**.

Pressor support (noradrenaline preferred over dopamine or adrenaline) as needed to maintain adequate mean arterial pressure.

Avoid nephrotoxic agents.

Continuous modes of **hemodialysis** if needed, such as venovenous hemodialysis.

6) Metabolic concerns

Follow closely: **glucose, potassium**, magnesium, phosphate.

Consider nutrition: **enteral feedings** preferred over total parenteral nutrition.

Paracetamol hepatotoxicity

N-acetylcysteine is well established in the management of an acetaminophen overdose and effective in preventing liver injury when administered **within 15 hours** of acetaminophen ingestion.

NAC is the antidote for paracetamol poisoning, **repleting glutathione that detoxifies the harmful metabolite**.

7) Liver transplantation

Hepatic transplantation has to be considered for patients reaching grade 3 and 4 coma due to acute liver failure.